



Large Language Models in human resource management: A systematic literature review of applications, open issues and future research directions

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ABSTRACT

The rapid adoption of Artificial Intelligence (AI), particularly Large Language Models (LLMs), in Human Resource Management (HRM) is transforming core functions including recruitment and selection, training and development and performance evaluation. Although prior studies provide conceptual discussions, empirical findings and methodological debates on AI in HRM, existing reviews largely examine AI at a broad level and lack a unified synthesis of LLM-specific contributions, risks and governance implications. To address this gap, this paper presents a systematic literature review of LLM-enabled HRM that integrates both technical and theoretical perspectives into a common analytical framework. Specifically, we classify the retrieved literature along two intersecting axes: (a) the technical footprint of LLMs (discriminative versus generative models) mapped to concrete HRM functions and (b) the theoretical lenses underpinning LLM adoption and use across five major research streams. Based on this synthesis, we identify recurring limitations, unresolved tensions and open issues that require deeper context-sensitive investigation. We further provide governance check-lists for practitioners, researchers and regulators, alongside a four-lane roadmap and a six-point future research agenda. Overall, the review contributes a structured and LLM-specific conceptualization of AI-enabled HRM that connects technological capabilities, organizational implications and governance requirements. Converting LLM promise into trustworthy and equitable HRM practice will require co-evolving robustness metrics, layered human oversight and harmonized audit standards.

1. Introduction

The digital era in which we live has profoundly reshaped the landscape of human resource management (HRM), especially by driving a shift from administrative and operational activities to data-driven strategic functions. Today, companies are facing the growing intricacies of globalization, remote employment, diverse workforces and fast-paced innovation. Consequently, HRM is increasingly compelled to implement digital tools to increase efficiency, adaptability and employee engagement (Budhwar et al., 2023). Studies suggest that when supported by involved work systems and transformative leadership, digital transformation enhances the resilience of both organizations and employees (Nguyen et al., 2024). Recent research has also highlighted the urgent need for HR professionals to develop digital fluency and technological sophistication, thus positioning HRM as a critical player in the wider digital transformation of organizations (Malik et al., 2022; Lu et al., 2025). This change demands not just the implementation

of new tools, but also a reevaluation of HR's role in fostering value, driving innovation and influencing organizational culture amid digital upheaval, among other areas.

Artificial intelligence (AI) has emerged in the last couple of years as one of the most powerful enablers of digital transformation in the HRM domain. Growing evidence highlights its ability to automate and enhance a wide range of HR-related functions in businesses and corporations. For example, AI-powered tools, including chatbots, virtual assistants and predictive analytics platforms, are all playing an expanding role in key HRM functions such as recruitment, onboarding, employee engagement, learning and performance management (Budhwar et al., 2023; Zhang and Hao, 2025). It is suggested that AI can increase efficiency and reduce bias in different HR functions while strengthening boundary expansion, agility and adaptability, as long as there is adequate perceived organizational support (Antoniadis et al., 2024; Ma et al., 2024). From a technical point of view, conventional

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HR “AI agents” automate a single well-defined task like, for example, screening a resume, answering a policy query and triggering an application programming interface (API) call. On the other hand, agentic AI orchestrates a constellation of such agents that share memory, re-allocate sub-tasks and pursue an end-to-end HR objective without human micromanagement and/or supervision (Sapkota and Roumeliotis, 2025). This distinction (AI agents vs. agentic AI) is crucial, since coordinated autonomy of agentic systems transforms siloed payroll systems into unified talent intelligence platforms where predictive workflows recruit, onboard, upskill and retain employees, thus delivering personalized experiences while stripping cost, friction, and bias from routine HR processes (Kshetri, 2025). The same necessity that makes agentic AI strategic also elevates the governance stakes and the adoption of AI in HRM is not without challenges. For example, bias mitigation, data protection and transparent oversight must now be designed for the orchestration layer itself, not only for individual task agents (Sapkota and Roumeliotis, 2025). In this vein, scholars have raised concerns about ethical risks, algorithmic bias, explainability and the potential dehumanization of HR processes (Budhwar et al., 2023; Rabenu and Baruch, 2025).

In the broader landscape of AI, large language models (LLMs), including discriminative and generative systems, represent a particularly transformative innovation and groundbreaking advancement for HRM. In general terms, discriminative LLMs are mainly oriented towards analyzing, classifying, matching or predicting outcomes from existing data, while generative LLMs are designed to produce new textual or conversational outputs, such as job advertisements, interview questions, feedback reports or chatbot responses. LLMs can facilitate natural language comprehension, dialog creation, text summarization, sentiment analysis and knowledge extraction, thus presenting new opportunities to automate complex, language-intensive HRM tasks (Campion and Campion, 2025; Budhwar et al., 2023; Nyberg et al., 2025). Models like ChatGPT (Generative Pretrained Transformer) can serve multiple functions acting as virtual HR assistants, conversation partners and content producers, while discriminative models excel in advanced classification, matching and predictive tasks (Raman et al., 2024). The significance and relevance of LLMs in HRM lies not only in their technical sophistication, but also in their ability to fundamentally transform how HRM professionals interact with data, systems, and, most importantly, with employees (Rabenu and Baruch, 2025). Whereas traditional HRM research has relied heavily on structured quantitative approaches, the emergence of generative and discriminative LLMs has unveiled unparalleled opportunities to automate, improve and rethink HR processes, ranging from recruitment and selection to employee development, performance management, and workforce planning.

Several authors have recently begun to assess the impact of generative AI (GenAI) tools, particularly ChatGPT and related conversational agents, on HRM-related functions. For example, Sakib et al. (2024), Bernik and Āprajc (2025) and Li and Cheng (2025) provide an early critical review focusing specifically on the benefits and challenges of AI chatbots (like ChatGPT) in HRM, further highlighting their potential to improve recruitment, on-boarding, employee support and engagement. The authors in Budhwar et al. (2023) and Jiang et al. (2025) expand this discussion by addressing the broader organizational, societal and ethical implications of GenAI in HRM, especially emphasizing both the promise and the profound risks these technologies introduce. Similarly, the authors of Rabenu and Baruch (2025) advocate a cyborg perspective arguing that AI does not just augment existing HRM practices but has the potential to fundamentally redefine the role of HRM, calling for a radical reconceptualization of the HRM value proposition in the age of LLMs. Other review papers cover LLM-related applications in specific HRM domains such as hiring decisions, recruitment applications or personality assessment in personnel selection (Tripathi et al., 2026; Trovão et al., 2025; Dunlop et al., 2025). Empirical work has also emerged to assess the performance

of specific generative LLMs (such as ChatGPT and Bard) on HRM-relevant tasks. Raman et al. (2024) present one of the first systematic benchmarking studies showing that while LLMs demonstrate promising precision and clarity on HR certification questions, their limitations, including hallucination risks and ethical concerns, require robust human oversight. Complementing these performance assessments, Rahman and Singh (2025) explore how GenAI can support organizational learning, offering opportunities for virtual coaching, personalized development and knowledge management, while also raising new challenges related to data quality, interpretability and ethical governance. Beyond generative systems, the authors in Campion and Campion (2025) provide a comprehensive review of text analysis methods in HRM, covering categorization, dictionaries, supervised and unsupervised Machine Learning (ML). Finally, Kimura (2023) underscores the critical need for ethical frameworks and methodological rigor when using LLMs to assess sensitive constructs such as personal values, underscoring the risks of bias and discrimination inherent in automated analytics.

Despite these important contributions, the existing literature remains fragmented, with conceptual papers focusing on theoretical possibilities, empirical works addressing narrow performance benchmarks and methodological reviews offering broad overviews of text analysis. More importantly, current research often treats AI in HRM as a general technological category and does not sufficiently capture the distinct role of LLMs as interactive, language-generating and decision-support systems that increasingly shape recruitment, evaluation, training and employee communication processes. This creates an important conceptual tension, since HRM is not only concerned with efficiency and automation, but also with fairness, trust, transparency, accountability and human judgment. Thus, there is a pressing need for a structured and systematic review that integrates discriminative and generative LLM applications across HRM functions, while synthesizing evidence on technical implementation, platform integration, human-in-the-loop (HITL) processes, reported limitations, open issues and practical implications. This review aims to address this gap by systematically analyzing recent research on LLM applications in HRM and by developing a comprehensive framework that not only maps the current landscape of LLM use in HRM, but also identifies critical opportunities, governance challenges and research frontiers for advancing both HRM practice and scholarship in the era of intelligent automation. Specifically, our research focuses on shedding light on the following Research Questions (RQs) that seem pertinent towards capturing the current state-of-the-art LLM applications in HRM:

- RQ1: What types of LLMs (generative or discriminative) are currently applied across specific HRM functions?
- RQ2: What are the key technical, computational or theoretical approaches underpinning current research on LLM applications in HRM and how do they shape the design and evaluation of these systems?
- RQ3: How do HITL mechanisms and ethical design considerations (such as bias mitigation, fairness and transparency) influence the trustworthiness, governance and deployment challenges of LLM-powered HRM systems and what open issues remain?

The remainder of this paper is structured as follows. Section 2 provides a theoretical background on LLM architectures, clarifying the distinction between generative and discriminative models and their relevance in HRM tasks. Section 3 outlines the methodology drawn for the review, describing the selection criteria, the search strategy and the content analysis process adopted. Section 4 presents a bibliometric overview of the reviewed literature, highlighting publication trends and thematic focus areas. Section 5 organizes the retrieved studies based on LLM-oriented technical applications in key HRM functions, while Section 6 provides a theoretical classification of the retrieved literature. Section 7 presents the main limitations of the available literature along with open issues relevant to the next generation of LLM-centric agentic tools in HRM. Section 8 synthesizes the main findings, offering a consolidated road map for future research directions. The paper ends with some concluding remarks provided in Section 9.

2. Theoretical background on LLMs: The distinction between generative and discriminative utility

Since the scope of this research is oriented towards mapping the main HRM objectives addressed in the current body of knowledge through the application of LLMs, the distinction between generative and discriminative AI approaches becomes essential. This differentiation forms the basis for the analytical framework used in the study and is applied consistently throughout the sample analysis. To support a clearer understanding of the key themes discussed in the following sections, this part of the paper provides a brief overview of the theoretical background of LLMs, focusing specifically on their implementation strategies to clarify whether they operate within a generative or discriminative AI paradigm.

LLMs are advanced AI systems designed to understand, generate and manipulate human language at scale. Practically, LLMs were made available for the first time in human history in 2017, when the seminal work of Vaswani et al. (2023) laid the foundations for the Transformer architecture. These models are based on deep learning architectures, most notably the Transformer architecture, which allows them to process vast amounts of text data and learn complex linguistic patterns. At their core, LLMs are trained on massive corpora of text using unsupervised or self-supervised learning objectives, such as predicting the next word in a sequence (causal language modeling) or filling in missing tokens (masked language modeling). The fundamental building block of an LLM is the attention mechanism, which enables the model to weigh the relevance of each word in a sequence relative to others, thus capturing contextual dependencies regardless of distance. The architecture typically consists of multiple stacked layers of self-attention and feedforward neural networks, allowing the model to build hierarchical representations of meaning. As the number of parameters increases by billions, LLMs gain the ability to perform a wide range of language tasks—translation, summarization, question answering, code generation and more—without the need for task-specific tuning.

A recent survey of the field underscores how quickly LLM research has outperformed scholars' ability to track it. Raiaa et al. (2024) synthesize the explosion of work since GPT-3 by mapping the full pipeline, from transformer fundamentals and token-level pre-processing, through scaling laws and hardware demands, to domain-specific deployments in health, education, business and agriculture, while also cataloging unresolved issues around privacy, bias, energy use and evaluation. Their taxonomy and historical timeline provide a concise backdrop for the present study. In particular, they show that LLM progress has been driven as much by engineering choices (data curation, parameter-efficient tuning, distributed training) as by algorithmic novelty and they highlight precisely the open technical and governance questions that our paper now revisits in the HRM context.

Implementation-wise, LLM-centric applications commonly employ a transformer neural network architecture. These architectures are typically categorized into three variants: encoder-only, decoder-only and encoder-decoder, each optimized for different types of language tasks. Encoder-only models, such as BERT (Bidirectional Encoder Representations from Transformers), focus on understanding the input sequence through bidirectional attention mechanisms, making them well-suited for tasks like text classification, named entity recognition and semantic similarity analysis. Decoder-only models, exemplified by GPT, utilize auto-regressive decoding to generate text one token at a time, which is particularly effective for tasks involving language generation, such as text completion, dialog systems and creative writing. Encoder-decoder models, such as T5 (Text-To-Text Transfer Transformer) and BART (Bidirectional and Auto-Regressive Transformers), integrate both components. The encoder processes the input to derive a comprehensive semantic representation, while the decoder generates an output sequence conditioned on this representation. This architecture excels in sequence-to-sequence tasks such as machine translation, abstractive

summarization and question answering. The selection of the appropriate architecture is critical in the LLM-centric design, as it directly influences the model's ability to comprehend, generate, or transform natural language effectively.

The recent advent of ChatGPT has revolutionized public and enterprise engagement with GenAI, yet it has also contributed to a blurry landscape regarding which applications truly qualify as GenAI-driven. This ambiguity extends to the development models behind such applications, particularly in relation to how LLMs are accessed and deployed. A key distinction lies between decoder-only models, like GPT-based systems, which are predominantly trained, hosted by large organizations and offered through API-based access and encoder-decoder models, which are more amenable to in-house fine-tuning and deployment. Decoder-only models are typically distributed via cloud-based APIs (e.g., OpenAI's GPT, Anthropic's Claude), allowing developers to integrate powerful language capabilities without managing the underlying infrastructure. This approach favors rapid prototyping and scalability, but limits transparency and customization due to proprietary constraints. In contrast, encoder-decoder models such as T5 or BART are often open-sourced and allow for more direct access to model weights, making them suitable for private deployment in secure environments, particularly when fine-tuned on domain-specific data. This flexibility supports applications that prioritize data privacy, cost control and tailored performance. As a result, the development strategy, AI-based versus in-house, significantly shapes the architecture, compliance and operational scope of LLM-centric applications.

As mentioned above, the field of HRM has increasingly attracted interest in the context of LLM-centric applications, particularly due to the growing need for intelligent, scalable and adaptive systems that support decision-making in dynamic organizational environments. GenAI can serve multiple HRM objectives, including automated talent acquisition through resume screening and candidate profiling, internal mobility recommendations, personalized learning and development pathways and even the simulation of realistic interview dialogs. However, to conceptually differentiate this evolving landscape, it is essential to distinguish between applications that merely augment existing HR processes using language-based automation and those that embody true generative intelligence—systems capable of producing novel insights, adaptive strategies or human-like interactions. Fig. 1 classifies only those large-language models that are actually cited in HRM-focused research—grouped by core architecture (encoder-only, decoder-only, encoder-decoder) and tagged as API-based or self-hosted—so it should be read as a literature-driven snapshot rather than an exhaustive catalog of all LLMs.

The evolutionary tree of LLMs, while primarily taxonomic, implicitly conveys a chronological progression that reflects the rapid advancement of the field. Foundational models such as BERT (2018) and GPT-2 (2019), situated near the base of the diagram, represent early milestones in discriminative and generative architectures, respectively. Subsequent branches illustrate the emergence of more advanced and specialized models—such as RoBERTa (2019), T5 (2019), GPT-3 (2020), GPT-3.5 (2022), and GPT-4 (2023)—signaling a temporal shift towards increased generative capabilities and scalable deployment. The introduction of Claude and Bard in 2023, followed by Gemini in 2023 and the open-source Mistral models in 2023 and 2024, marks the latest phase of innovation. This upward and outward branching mirrors the historical trajectory of LLMs, from research-driven, encoder-based models to powerful, API-based generative systems, capturing both architectural evolution and the growing industrialization of AI.

Apart from their technical design, LLMs are currently acting as socio-technical systems impacting organizational decision-making, leadership practices and long-term workforce sustainability (Albanna et al., 2025). For example, LLMs can support organizational learning by helping people find patterns, generate ideas and understand complex information. They can also help teams to share knowledge, discuss

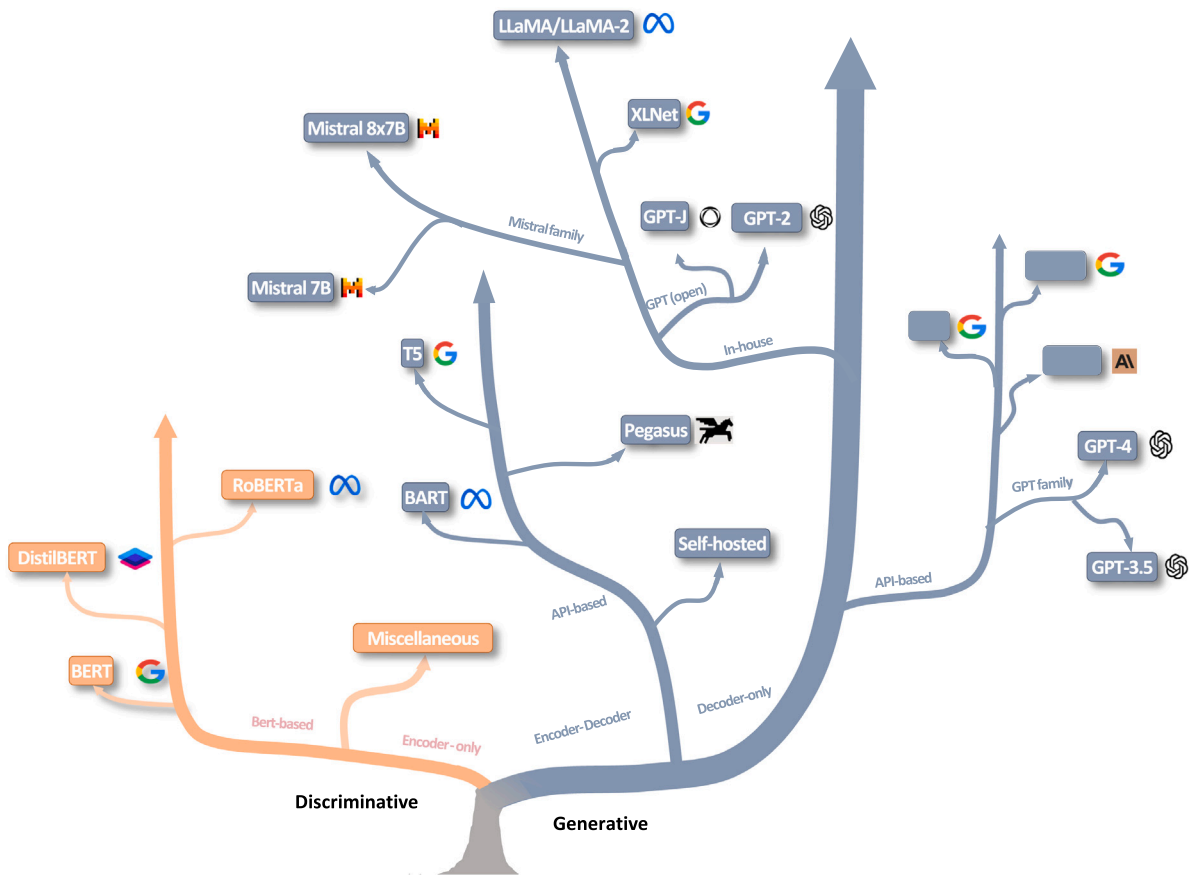


Fig. 1. Taxonomy of LLMs by architecture and deployment mode found in HRM-related literature.
Source: Adapted from Yang et al. (2023).

problems and create common understanding across the organization. In this way, LLMs act as a tool that connects individual learning, team learning and organizational knowledge, although human judgment and collaboration are still essential (Lenart, 2025). In HRM environments, LLMs are not mere computational tools for automating isolated tasks, but components of wider decision-support ecosystems that can affect recruitment strategies, employee evaluation, organizational learning and managerial communication. This perspective is in line with recent research grounded on the principles of a digital circular economy which stresses that digital transformation and sustainability depend not only on technological infrastructures but also on human-centered organizational practices and sustainability-oriented HRM systems (Singh et al., 2025a,c). Empirical evidence from manufacturing firms also shows that HRM practices such as recruitment, training and performance appraisal affect the adoption of digital circular economy models and organizational sustainability outcomes (Singh et al., 2025b). It is, therefore, clear that understanding the distinction between discriminative and generative LLMs is important not only from a technical perspective, but also from an organizational and governance perspective, as different architectural choices may influence explainability, accountability, adaptability and trust in LLM-supported HRM systems.

Recent labor-market evidence further suggests that the impact of LLMs depends not only on what these systems can theoretically do, but also on how extensively they are actually deployed in organizational workflows. A recent large-scale analysis combining task-level AI capability estimates with real-world usage data finds that current AI deployment remains substantially below its theoretical potential, although occupations with higher observed AI exposure already show weaker projected employment growth and early signs of slower hiring among younger workers. Particularly exposed occupations include customer service, financial analysis, data-entry and administrative work, all domains closely related to HRM-supported organizational

processes (Massenkoff and McCrory, 2026). These findings reinforce the importance of examining LLM adoption not merely as a technological issue, but as a broader organizational and workforce transformation challenge.

3. Methodology

For conducting our review, we have relied upon a systematic methodological approach. Specifically, the designed approach encompasses five methodological phases in order to implement a fully transparent and reproducible research methodology, which aligns with the methodology proposed by Briner and Denyer (2012) while incorporating the core elements of the PRISMA statement (Moher et al., 2009) to conduct systematic reviews of the literature. The core of the methodology is presented in Fig. 2.

Arguably, to conduct a review study, certain factors must be generally met: (a) the research topic attracts significant scholarly interest in the pertinent literature, particularly in recent years, and (b) the volume of the underlying literature is sufficient to yield adequately generalizable conclusions (Munn et al., 2018). The topic centered at the intersection of HRM functions and LLM-driven applications and theoretical concepts presents a particularly compelling case. A notable finding is that, despite the relative novelty of the field, a considerable number of theoretical or survey-like papers have already emerged. However, a closer inspection reveals that the overall research activity and publication output in this domain have been largely concentrated within the past five years, with the most substantial volume of contributions published during the last three years. This temporal concentration is consistent with the broader development timeline of LLMs, as the transition from transformer-based and pretrained language models to

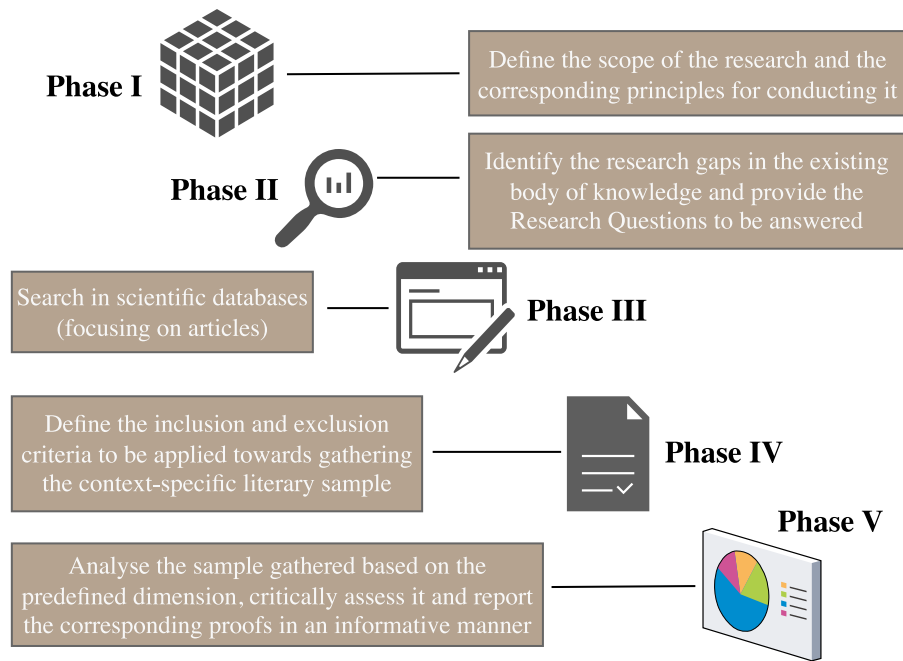


Fig. 2. The overall methodology adopted for conducting the review.

widely accessible generative systems has accelerated their scholarly and practical uptake in organizational settings. A significant milestone in this timeline could be regarded as the public online release of ChatGPT in late 2022, which arguably marked a major turning point in the visibility, accessibility and organizational experimentation with generative LLMs, helping to explain the sharp increase of HRM-oriented LLM studies in the post-2022 period (Budhwar et al., 2023; Sakib et al., 2024). This observation suggests a rapidly growing scholarly interest, yet also reflects a field still in its formative stage.

This pattern underscores the need for a more granular, systematically structured review, particularly one that maps the specific HRM objectives addressed through LLM-centric approaches. Unlike umbrella reviews that synthesize well-established fields, a systematic review is more suitable for an emerging and fragmented field, as it allows for the classification of studies, identification of thematic gaps and detection of nascent trends. It also supports the development of shared terminology and conceptual clarity, which are essential steps towards consensus building. Therefore, given the current state of the literature, a focused systematic literature review not only aligns methodologically with the maturity level of the field but also serves a strategic function in shaping future research directions and informing practical adoption of LLMs in HRM.

Since the scope of this review is tailored to analyze LLM-specific applications within the HRM domain and to highlight underlying research gaps, we strategically define our core sample to support an in-depth structured analysis. Consequently, our approach is based on a carefully curated selection of 141 peer-reviewed papers that directly address the intersection of LLMs and HRM functions. Given the emerging and relatively fragmented nature of this field, our study is positioned within in-depth mapping reviews. Elaborating upon our previous argument that LLM-centric applications can manifest in both generative and discriminative forms, our search strategy was designed to capture this duality by combining terminology related to both HR-specific functions and LLM-related technologies. To identify relevant publications, we performed systematic searches in major databases of the scientific literature, including Scopus, Web of Science, and IEEE Xplore, without imposing any time-frame restrictions. To identify context-specific keywords for searching, we have consulted previous research conducted within the GenAI topic (Budhwar et al., 2023) and the broader topic

of ML (Garg et al., 2022), as well. Although generic ML approaches are generally out of the scope of this research, analyzing previous research efforts facilitates a sound definition of HRM-specific keywords and classes.

Building on these domain-specific insights, we constructed a comprehensive search strategy integrating HRM-related terms (e.g., “human resource”, “recruitment”, “talent management”, “workforce analytics”) with LLM terminology (e.g., “large language model”, “transformer”, “GPT”, etc.). We also reviewed the search strategies used in related literature reviews to refine our approach and ensure coverage of both conceptual and applied dimensions. During the data collection phase, we applied these combined keywords using the advanced search features available in each database, focusing on their presence in article titles, abstracts or indexed keywords. To support the transparency and reproducibility of our search process, we include below the query formulated for the Scopus database.

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TITLE-ABS-KEY (("generative artificial intelligence" OR "large language model*" OR "LLM*" OR "discriminative artificial intelligence" OR "transformer model*") AND ("human resource management" OR "HRM" OR "recruitment" OR "talent acquisition" OR "resume screening" OR resume OR "CV screening" OR "candidate screening" OR "personnel selection" OR "personnel attrition" OR "employee attrition" OR "personnel turnover" OR "employee turnover" OR "employee retention" OR "workforce planning" OR "succession planning" OR "personnel training" OR "personnel development" OR "learning and development" OR "competency management" OR "skills assessment" OR "employee performance" OR "personnel performance" OR "personnel compensation" OR "reward management" OR "payroll management" OR "personnel attendance" OR "employee attendance" OR "personnel health" OR "personnel safety" OR "occupational health"))
  
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3.1. Inclusion and exclusion criteria

The search process for collecting the literature sample was conducted in May 2026. After completing the search across the three scientific databases, we collected a final sample of 615 articles. It is worth noting that, although some of these papers were initially identified through multiple databases, including Web of Science and IEEE Xplore, all selected contributions were also available in the Scopus database. In addition, we focused exclusively on papers published in peer-reviewed journals. This approach minimizes the introduction of “noise” into our analysis and aligns with established methodologies employed in previous systematic reviews (Snyder, 2019). It also served as a baseline quality-control criterion for the core sample, without replacing a formal risk-of-bias or methodological-quality appraisal of each individual study.

The eligibility of the collected study sample was meticulously evaluated by applying a three-fold forward approach consisting of title screening, abstract screening, and full-text reading. In the first stage of the three-fold selection process, we reviewed the titles of all collected articles as an initial assessment of their relevance to the scope of our review. The application of this first-level eligibility criterion resulted in the exclusion of 291 articles. The main reasons for exclusion during this phase included: (a) lack of thematic relevance to HRM core functions (a significant proportion of excluded papers employed LLMs or other AI techniques in adjacent or unrelated domains, or dealt with broader organizational dynamics such as corporate learning transformation, skill gap analyses, or institutional adaptation to AI), (b) method-centric or technical papers without HR deployment (papers that proposed novel methods other than LLMs, such as natural language processing and large-scale text mining, without a clear HRM application); and (c) conceptual, opinion-based, or normative analysis papers that did not cover aspects of the usage of LLMs in HRM (e.g., the future of HRM and/or the ethics of AI).

In the second stage of the screening process, we reviewed the abstracts of the remaining eligible articles. This process led to the exclusion of 124 articles. In the third stage of eligibility assessment, we conducted a full reading of the remaining two hundred (200) articles. At this stage, fifty-nine (59) studies were excluded. The articles excluded in the last two stages were found to be pertinent to topics within the broader domain of GenAI, but their focus did not align with our target scope on LLM-centric applications. For example, several studies concentrated on generative adversarial networks (GANs), which, despite their relevance to the generative paradigm, fall outside the architectural and functional boundaries of LLMs. Following the above approach, we finalized the collection of one hundred forty-one (141) relevant articles, completing the forward search process. To increase the transparency and reproducibility of our approach, the interested reader is referred to Fig. 3, which provides the detailed steps we followed to achieve content saturation.

3.2. Content analysis and synthesis

Thematic content analysis provides a robust methodological framework for systematically examining qualitative data, enabling researchers to identify, scrutinize and interpret patterns of meaning, commonly referred to as “themes”, within qualitative datasets (Dasaklis et al., 2022; Elo and Kyngäs, 2008). In this study, we have meticulously adopted a thematic content analysis approach to distill research areas and discern overarching themes from the corpus of the retrieved literature. Using content analysis techniques, we facilitate the extraction of specific qualitative attributes, which subsequently enables the grouping of articles that share common characteristics through content synthesis

methods. To systematically extract relevant information from the selected instances, content analysis was performed within the following coding themes:

1. Contextual and functional scope

- HR function and objective: The specific HRM function addressed (e.g., recruitment, employee evaluation) and the intended purpose (e.g., automation, decision support, personalization).
- Type of contribution: Classification of the study’s focus—conceptual analysis, computational experimentation, background synthesis, ethical discussion, prompt engineering, statistical-type hypothesis testing, or technical implementation.

2. Technical and architectural implementation

- Deployment model: Indication of whether the approach is API-based (e.g., ChatGPT, Gemini) or in-house trained using open-source LLMs.
- Model scope and typology: Identification of the approach as generative or discriminative and description of any integration with traditional machine learning techniques.
- Integration with other technologies: Interfaces with broader infrastructures such as Big Data pipelines, cloud platforms or enterprise systems.

3. Design, governance and interaction mechanisms

- HITL design: Presence of predefined processes that incorporate human oversight, feedback or decision-making into the system lifecycle.
- Formal representation and classification: Thematic structuring of content, enabling analytical categorization of concepts, technical patterns, and application contexts.

Upon collating the qualitative data, narrative-oriented techniques were applied to synthesize the extracted information and create context- and methodological-specific classes, helping to facilitate our research scope. Narrative synthesis is a respected approach that focuses on elucidating how studies, which explore different aspects of the same phenomenon, can be coherently summarized to construct a holistic understanding of that phenomenon (Campbell et al., 2019). To ensure maximum consistency in our research and effectiveness in identifying critical themes in the retrieved literature, we extend previous approaches presented in the literature (i.e.; Dasaklis et al., 2024) by adopting a novel, yet actionable scheme that integrates automated content-analysis tools with human oversight. Specifically, after collecting the articles and deciding on the thematic analysis dimensions (coding), two independent authors analyzed the selected literature, one using a predefined prompt via an LLM-based chat interface, and the other employing dedicated software for thematic content analysis (MaxQDA 2024). Since the literature analysis was conducted in two successive stages, the same predefined query was applied in both stages: during the first stage of the analysis, conducted in August 2025, the prompt was implemented through ChatGPT-4o, while during the second stage, conducted in May 2026, it was implemented through ChatGPT-5.2 or the most recent ChatGPT model available at the time of the analysis. For facilitating the reproducibility of our work, we also provide the prompt made by the first independent researcher:

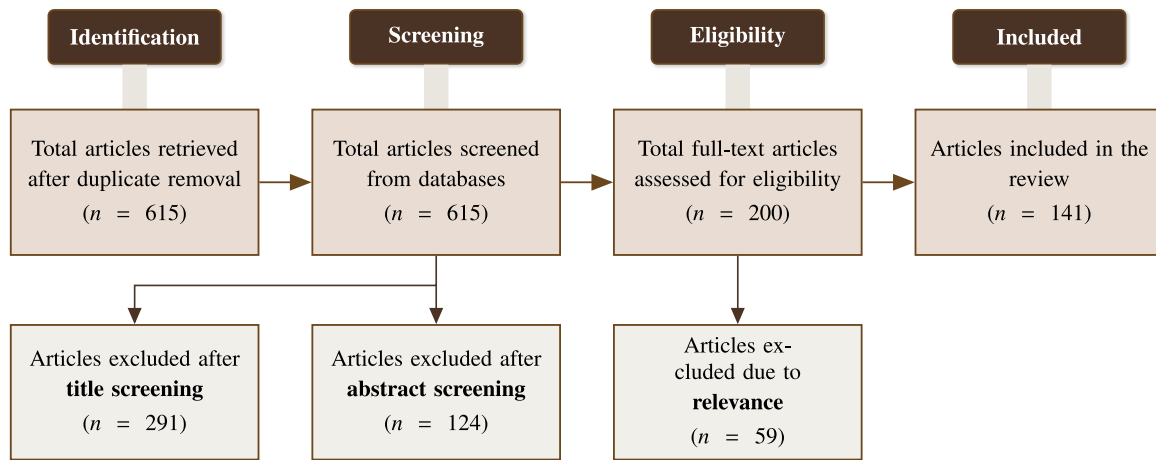


Fig. 3. Screening and selection process of research articles.

LLM-assisted coding prompt

Carefully read the attached article and extract only explicitly reported information according to the following dimensions:

(1) what is the main objective of the paper and to which HR function is it associated? Based on the experimentation and the structure of the paper, in which of the following classes would you add it? The possible classes are: [conceptual analysis, computational experimentation, background synthesis, ethical discussion, prompt engineering, statistical-type hypothesis testing, or technical implementation.];

(2) provide details regarding the technical and architectural implementation, including the deployment model, model scope and typology, and integration with other technologies; and

(3) beyond the implementation, also provide details on the design, governance and interaction mechanisms, including HITL design and formal representation/classification.

Do not infer information that is not explicitly reported in the article.

If a coding element is absent, unclear, or not sufficiently documented, mark it as “not reported” or “unclear” and briefly justify the classification based only on the article text.

This procedure was performed separately for each article and in parallel with human oversight, forming a triangular validation approach aligned with the recent protocol proposed by Jensen et al. (2025) for applying ChatGPT-based support to systematic review data extraction when its outputs are validated against human reviewer decisions. In our case, this triangulation process was used to collect the first coding through the LLM-assisted route, while the human-based analysis provided an independent coding route. Before moving to the next paper, the first researcher assessed whether the LLM-assisted coding results were reasonable in comparison with the initial screening and full-text inspection already conducted. When the output for specific coding variables was considered incomplete, targeted follow-up prompts were used, while the final coding decision remained subject to researcher judgment. To assess the degree of convergence between the two analyses, we applied the Holsti coefficient (see Eq. (1)), which yielded a value of 0.84; thus, reflecting a strong consensus and suggesting that the coding process was quite consistent and reliable.

$$\text{Holsti Coefficient} = \frac{2 \times n_{\text{agreement}}}{N_{\text{coderA}} + N_{\text{coderB}}} \quad (1)$$

4. Bibliometric analysis

The final corpus of literature reviewed in this study comprises a total of one hundred and forty one (141) papers focusing on the application of LLMs in HRM. Of these, sixty-three (63) papers were classified

as theoretical or conceptual, including works that propose frameworks, explore normative questions, illustrate use cases, or critically reflect on the integration of LLMs into HR functions. The remaining seventy-eight (78) papers fall into the category of technical contributions, characterized by the implementation, evaluation, or computational analysis of LLMs in practical HR contexts, such as recruitment automation, performance evaluation, and training personalization, among others. This division allows for a dual-layered examination: one that captures the evolving conceptual discourse around LLMs in HRM and another that assesses the technological maturity and functional capabilities demonstrated across applied studies. Each category is analyzed in its own dedicated section, allowing for a balanced overview of both the theoretical foundations and the operational realities of the adoption of LLM in HRM.

Fig. 4 visualizes the keyword co-occurrence network for our corpus and reveals how scholarship on LLMs in HRM is currently organized around three loosely connected thematic clusters. The first, centered on terms such as “generative ai”, “human resource management”, “ChatGPT”, “recruitment”, “resume screening”, “training”, “job matching” and “candidate”, captures application-oriented work that adapts GenAI tools to core HRM workflows. A second cluster, organized around “large-language model”, “llm”, “bias”, “fairness”, “prompt engineering”, “gpt4”, “llama” and “synthetic data”, reflects emerging concerns with LLM design, governance and responsible deployment. A third group, focused on “natural language processing”, “BERT”, “dataset”, “performance”, “accuracy”, “classification” and related embedding-based terms, points to method-driven studies that refine model architectures or benchmark discriminative tasks. Bridging terms such as “ChatGPT”, “job matching”, “efficiency” and “performance” sit at the interfaces of these clusters, signaling a shared concern with both operational gains and the constraints of LLM adoption. Arguably, the map indicates a field that remains fragmented: tool-focused, application-specific and method-heavy, but increasingly oriented towards practical impact, evaluation and governance questions. The bibliometric mapping in Fig. 4 was generated using VOSviewer (van Eck and Waltman, 2010), applied via version 1.6.20 (released October 31, 2023).

By critically analyzing the bibliometric evidence and the temporal evolution of textual co-occurrences, it emerges that LLM-centric applications are gradually evolving from discriminative, back-end analytical applications towards generative, interactive and decision-support configurations embedded in HRM workflows. Earlier studies adopted LLMs mainly as discriminative tools, orienting their applicability towards classification, semantic matching, ranking and prediction tasks, broadly adopted for objectives including resume screening, candidate-job fit

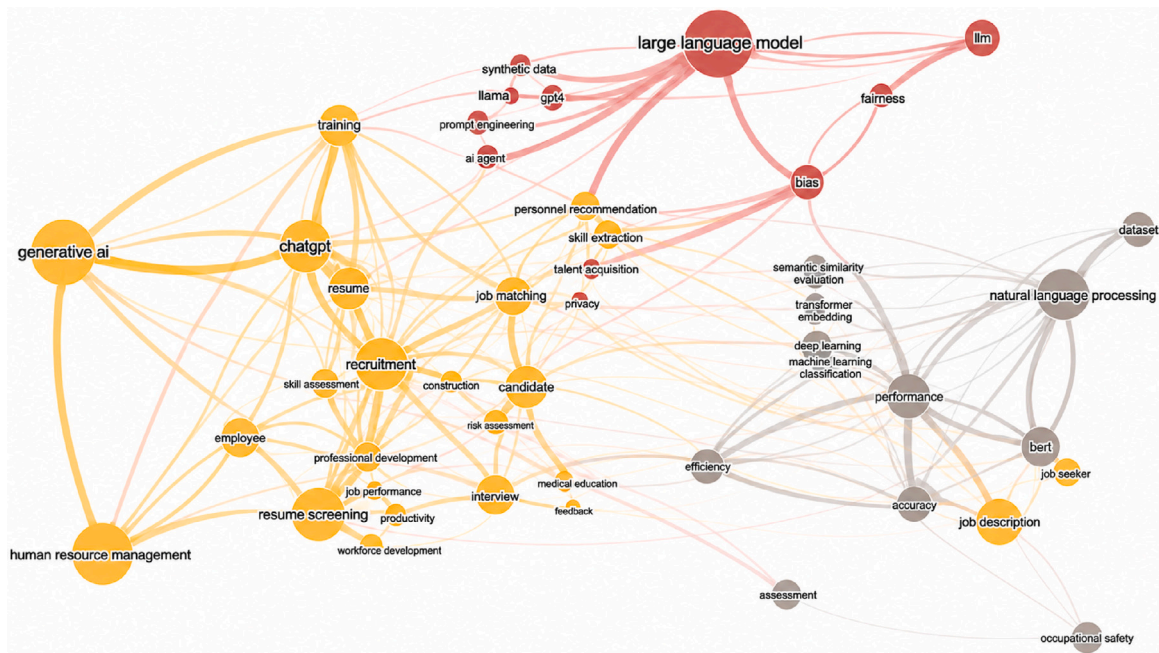


Fig. 4. Keyword co-occurrence network of LLM-HRM scholarship (2021–2026).

estimation, skill classification, workforce planning and attrition prediction, among others. The orientation of this phase is further supported by the evidence that almost two out of three articles published before 2023 build on encoder-only models, such as BERT and its variants, based on the elements presented in the following section. Common areas that gathered considerable traction in this phase include representation learning, text classification and similarity-based decision support tasks. In contrast, from 2024 onwards, the field shows a clear acceleration, with almost three out of four contributions in the overall sample appearing in this more recent phase. In this phase, the co-occurrence analysis indicates a stronger shift towards the integration of GenAI tools into organizational decision loops, with almost two out of three publications referring to at least one term associated with generative implementation, such as ChatGPT, GPT, GenAI, conversational agents, prompt engineering or synthetic content. From a technical perspective, significant research streams in this phase are associated with text generation, summarization, explanation, simulation and conversational support, broadly relevant to job-ad creation, interview support, onboarding, personalized training, performance-feedback preparation and HR analytics, among others.

5. Taxonomy of LLM-centric tools and applications in HRM

This section discusses the key contributions presented in the recent literature on LLMs in HRM. To facilitate a better understanding of the following synthesis, we also present Fig. 5, which identifies the main HRM functions addressed through LLM-centric approaches. This taxonomy serves as the backbone for formulating the evidence of this section and is developed through a two-level granularity scheme. At the first level, it presents the main HRM functions, while at the second level, it illustrates the major themes identified as research attractors within the topic under consideration.

As is evident, twelve (12) HRM functions have been identified in the assessed literature sample. Among them, *Recruitment and Selection*, *Training and Development*, and *HR Information & Decision Support Systems* have attracted the highest research interest. However, it should be noted that *HR Information & Decision Support Systems* is mainly observed as a cross-functional enabling layer rather than as a stand-alone HRM function. Several contributions have also been oriented towards

functions such as *Workforce Planning*, *Performance Management*, and *Occupational Health & Safety*. To better reflect the main thematic concentration of the literature and guide the reader through the evidence synthesis, the remainder of this section discusses the most frequently represented functions as stand-alone sub-sections, while functions with equal or fewer than five (5) contributions, as well as functions with only marginal representation in the technical sample, are grouped into a generic class, namely *Miscellaneous*.

A worth-mentioned aspect of the technical sample analyzed in this section emerges from the distribution of papers based on the function-specific objectives addressed. This distribution is presented in Table 1. Although the taxonomy identifies twelve (12) HRM functions, the table focuses on the functions that appear with sufficient recurrence either as stand-alone topics or in combinations with other functions. The evidence suggests that a significant proportion of articles address more than one HRM function within their proposed applications, thereby reflecting the increasingly integrated and cross-functional orientation of contemporary HRM research. The most commonly observed combination is that of *Recruitment and Selection* with *HR Information Systems/HR Analytics*, which appears in twelve (12) papers, indicating a prevailing emphasis on the data-driven optimization of recruitment processes. Other frequent combinations include *Recruitment and Selection* with *Workforce Planning*, appearing in seven (7) papers; *Recruitment and Selection* with *Training and Development*, appearing in six (6) papers; and *Training and Development* with *Health and Safety*, appearing in five (5) papers. Additionally, combinations such as *Training and Development* with *HR Information Systems/HR Analytics*, *Training and Development* with *Performance Management*, and *HR Information Systems/HR Analytics* with *Workforce Planning* are each recorded four or three times, respectively.

It is also important to note that some functions, including *Culture Management*, *Workforce Planning*, and *HR Information Systems/HR Analytics*, are not addressed as standalone topics in the technical sample, but appear in combination with other HRM functions. This suggests that these functions are typically explored as part of broader, multi-dimensional HRM initiatives rather than as isolated areas of investigation. This point is particularly important for *HR Information Systems/HR Analytics*, which should be interpreted less as an underexplored function and more as a cross-functional layer that supports other HRM

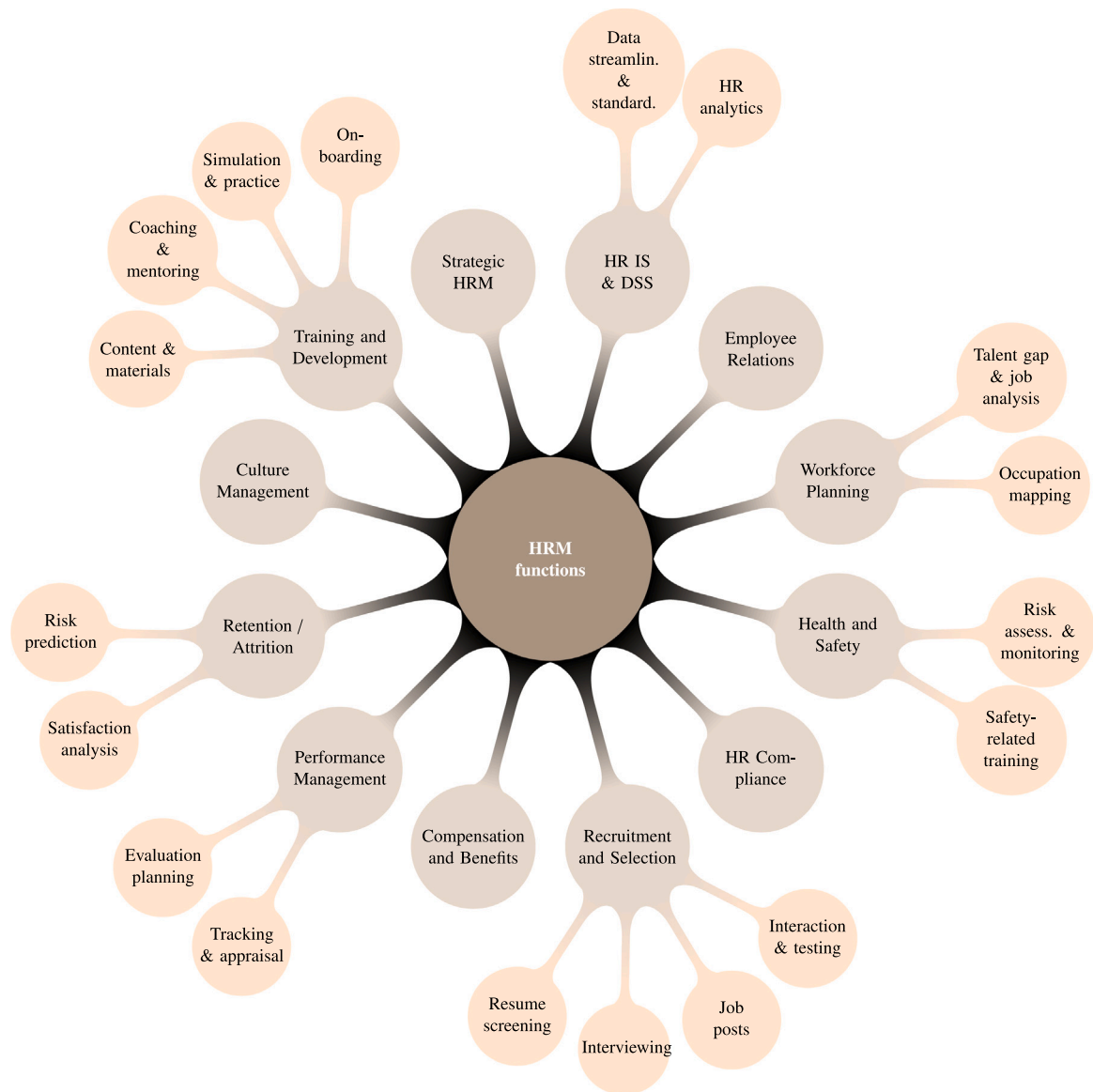


Fig. 5. A classification of LLM-based applications in several HRM functions.

Table 1

Frequency of HR Functions (stand-alone, in combinations, and total).

HRM function	Stand-alone	Total	Recruit. & Sel.	Train. & Dev.	HRIS/HR Anal.	Retent./Attrit.	Perf. Mgmt.	Workforce Plan.	Health & Safety	Culture Mgmt.
Recruitment & Selection	31	49	–	6	12	2	1	7	0	1
Training & Development	7	24	6	–	4	2	4	3	5	1
HRIS/HR Analytics	0	17	12	4	–	3	2	3	0	1
Retention/Attrition	1	6	2	2	3	–	2	2	1	1
Performance management/Evaluation	1	9	1	4	2	2	–	1	3	1
Workforce planning	2	9	5	3	3	2	1	–	1	0
Health and Safety	4	12	0	5	0	1	3	1	–	0
Culture management	0	1	1	1	1	1	1	0	0	–

applications. In the following content synthesis, papers addressing more than one HRM function are intentionally discussed in each relevant sub-section, towards validating the applicability of LLMs across the broader spectrum of HRM objectives. This approach reflects our aim to capture the common pathways through which LLM-centric applications are embedded in cross-functional HRM workflows.

5.1. GenAI applications

As previously identified from the collected sample, one out of two, specifically 56% of the articles present, introduce or extend an architectural paradigm or an end-to-end implementation scheme for incorporating LLM into managerial decision-making. Elaborating upon

our in-depth coding and the key distinctions in the implementation scope and the underlying architectures between GenAI and discriminative use of LLMs, we commend that from the articles presenting an application, about 52% of them rely on GenAI principles. In this sub-section, we consolidate the information regarding this part of the collected sample.

5.1.1. Recruitment and selection

The selection of personnel with the essential skill sets to meet job requirements has been a long-standing concern in HRM decision-making. In today's landscape, recruitment analysts face an increasingly complex environment shaped by rapidly changing job roles, an increasingly global talent pool and increasing pressure to balance speed, accuracy and fairness in hiring. They must also navigate large volumes of data and address ethical issues such as algorithmic bias and transparency. Beyond these challenges, recruitment decisions significantly influence the strategic direction and structure of an organization, creating multiple points of interaction with other HRM functions, such as talent development and training, that support long-term workforce planning and adaptability. Towards mitigating the complicated nature of such tasks, several GenAI applications have recently emerged. After an in-depth synthesis of the research findings, we could draw three streams of major interest within the GenAI applications dedicated to recruitment. In what follows, we discuss these streams, primarily by consolidating evidence regarding the implementation specifics.

The first stream refers to the automation of resume screening and candidate-job matching. Across this stream, studies move from keyword-based or rule-based filtering towards context-aware semantic alignment, where GenAI systems extract, summarize, complete, or compare candidate and job information. Integrated pipelines have been proposed to automate sentence classification, resume summarization, numeric grading and shortlisting decisions through LLM agents that simulate HR specialists (Gan et al., 2024). These implementations use both API-based models, such as GPT-3.5/4, and in-house or fine-tuned open-source models, such as LLaMA2. Related studies compare classical ML algorithms with LLM architectures for curriculum concordance and job-fitting tasks, reporting stronger semantic alignment in unstructured text, while also identifying challenges related to label mismatch and domain-specific ambiguity (Maree and Shehadeh, 2024). Synthetic-data strategies have also been explored, with fine-tuned GPT models generating labeled artificial resumes to augment real-world datasets and improve downstream classifiers such as BERT and feed-forward neural networks (Skondras et al., 2023).

A second group of studies extends resume screening towards generative recommendation and profile-enrichment mechanisms. Xu et al. (2026) propose the Adaptive Talent Aligner, an LLM-enhanced person-job matching architecture that combines dynamic hierarchical analysis, analytic hierarchy process, Redis-based preference and bias-correction memory pools, word2vec similarity and Ebbinghaus-style forgetting. The framework is benchmarked through ablation studies and comparisons with mainstream HRM systems in terms of recommendation accuracy, preference adherence and bias mitigation. Similarly, Zheng et al. (2026) introduce a generative job recommendation framework (GIRL), in which an LLM generates suitable job descriptions from candidate CVs, a reward model captures CV-job-description fit and proximal policy optimization aligns the recommender with recruiter-like preferences. This work is evaluated on two real-world datasets against seven baseline methods, together with GPT-3.5-turbo-based and human evaluation. From the opposite side of the candidate-job matching relation, Zhu et al. (2025) propose an LLM-based resume completion framework that generates missing or outdated resume content, especially recent work experiences, so that job-seeker profiles can better reflect current preferences in online recruitment platforms. The implementation follows a two-stage alignment process, including supervised fine-tuning on real resumes and Denoised Direct Preference

Optimization, and is evaluated through large-scale offline datasets and online A/B testing.

Beyond resume-related applications, the second stream of interest pertains to the interview stage of the recruitment process. This stage has traditionally been a focal point within the field, with core competencies centered on the evaluation of the skills sets of candidates, as well as the evaluation of behavioral attributes and personality traits. Recent GenAI applications have been tailored to both of these evaluation streams. From the evaluation of technical skills, the strongest concentration of evidence appears in IT and medicine, where standardized tasks and domain-specific rubrics make LLM evaluation easier to benchmark. In the IT industry, Szandala (2025) deploy ChatGPT-4, Google Gemini and Mistral 7b as “virtual technical interviewers”. Each model evaluates candidate responses using the same criteria as human DevOps reviewers, providing a fast and quantitatively comparable method for initial screening in technically structured recruitment contexts.

In medicine, GenAI has been examined mainly as a tool for benchmarking or augmenting high-stakes selection procedures. Tsoutsanis and Tsoutsanis (2024) assess Bing Chat, ChatGPT-3.5, Google Bard and LLaMA 2 on 100 medical multiple-choice questions from the Clinical Problem Solving section of a formalized medical exam and compare their performance with human candidates. In residency selection, Markow et al. (2024) examine a ChatGPT-4 instance that processes deidentified medical student performance evaluation letters and a detailed classification rubric, assigning applicant scores across multiple sessions and orders. These studies focus on rubric-based and text-bounded evaluation settings, where model outputs can be compared with established human-assessment procedures.

In addition to hard-coded technical skills, soft skills are important when it comes to recruitment. The assessment of soft skills and emotional intelligence of an applicant presents a focal point in both online and in-person reviews, in which the recruiter tries to inherently assess the behavioral traits of an applicant. Since we are moving to the new, remote-intended era, the use of GenAI protocols have been found to hold merit in terms of formalizing trait analyses, stemming from LLM-driven interactive mock-interview simulation tools capable of including adaptive follow-up questions (Daryanto et al., 2025). In this direction, Ghebatha et al. (2026) evaluate a zero-shot AI self-interview system for leadership competency assessment, where GPT-4o-mini and Qwen 32B generate interview questions, follow-up questions, behavioral insights and competency scores, while benchmarking relies on HITL expert ratings of insight quality, follow-up-question quality and answer-score fit. Similarly, Zhang et al. (2024) present a multi-stage LLM-based pipeline for assessing asynchronous video interviews, where GPT-3.5 and GPT-4 generate personality ratings, competency scores and overall performance ratings from transcribed responses, with outputs compared against human recruiter selections. At the same time, Phillips (2024) show that applicants can use generative LLMs to manipulate personality-test responses, achieving higher conscientiousness and integrity scores than most human test-takers, even though forced-choice items reduce this effect.

The third stream of GenAI literature is centered on job ad creation and labor-market text structuring. Research findings in the field indicate that the structure and phrasing of job postings substantially affect the qualitative attributes of the applicant pool. Consequently, the development of compelling and strategically aligned job advertisements constitutes a critical component of talent acquisition and has significant implications for the broader human resource planning of an organization. Studies in this stream use GenAI not only to generate or refine job advertisements, but also to transform unstructured job postings into structured and searchable representations. Chen et al. (2025b) propose an internal fine-tuned GPT-based agent with prompt tuning that parses 58,732 healthcare job postings into structured fields, such as title, responsibilities, skills and qualifications, with 95% extraction accuracy. Chumwatana and Hpone (2025) use GPT-4o to extract primary skills, tools and competencies from recent IT vacancies, cluster

synonyms and rank cleaned terms with TF-IDF. Similarly, Li et al. (2025b) introduce the LLM4Jobs tool, which uses an open-source LLM, such as Mixtral-8 × 7B, to remove non-essential company-culture text, summarize duties, embed job descriptions and ESCO/ISCO definitions, and retrieve the most similar occupation codes. These approaches connect job-ad processing with standardized skill and occupation representations that can be reused for screening, recommendation and workforce planning.

On-boarding, socialization and candidate preparation present some other worth-mentioning activities within the recruitment function. These activities are positioned in the interplay with other HRM functions. For instance, preparing candidates for interviewing process could be attributed to the Training process, although when it comes to resume preparation, this creates a multi-faceted approach. In this setting, LLM-based approaches have the capacity to analyze the resume, suggest alignment improvements for a target role, rewrite individual bullet points, generate probable interview questions and conduct a text-based mock interview with critiques of the student's answers (Flink et al., 2025). On the other hand, LLM-powered chatbots have the potential to create customized on-boarding support, especially when coupled with multi-criteria decision analysis methods (MCDA), such as the Analytic Hierarchical Process (AHP) (Agarwal, 2025). Both of these paradigms illustrate how recruitment-oriented GenAI tools may also support preparation, socialization and early employee guidance.

5.1.2. Training and development

The enablement of ongoing learning and the development of working capacity have been seen as central components of long-term HR development. In today's environment, with rapid technological progress and organizational change, training practitioners must create learning environments that can be adaptable, scalable, and tailored, in order to cope with fluid job requirements and shifting expectations from workers. At the same time, they must contend with the constraints posed by instructional quality, learner motivation, and efficient use of resources, in a manner that aligns with a wider talent development strategy. Against this backdrop, GenAI and specifically LLMs have emerged as useful tools that can extend and reshape training frameworks. From analyzing the relevant scholarly contributions of our sample, we identify three main domains for application of GenAI in training and development: (a) customized career coaching and alignment; (b) building discipline-specific and behavioral skills through interactive simulation; and (c) generation of pedagogical content and knowledge support for structured educational and workforce training environments through automation.

The first stream explores the use of GenAI as a virtual career assistant, supporting people through self-assessment, CV optimization, and targeted skill development. In this context, Flink et al. (2025) propose a multistep pipeline using GPT-4 to analyze user resumes, generate customized interview questions, revise bullet points, and evaluate job alignment based on role descriptions and sets of anticipated questions. The system also provides follow-up coaching and evaluative feedback in all sessions, establishing a closed-loop feedback mechanism that promotes continuous learning development. Although this stream intersects with recruitment, its training relevance lies in the provision of individualized feedback cycles that support the gradual development of career-related competencies.

In the second stream, GenAI tools are used to simulate real-world environments and provide adaptive training experiences. In the healthcare sector, Tran et al. (2024) integrate ChatGPT and DALL-E 3 to facilitate visual learning for nursing students through comic-style narratives, while Nofal et al. (2025) develop immersive metaverse-based interview simulations that merge VR environments with ChatGPT-powered avatars and multi-modal feedback. In a similar healthcare simulation setting, Li et al. (2026) introduce AutoCLC, an automated assessment and feedback system that processes audio recordings from Advanced Cardiovascular Life Support simulation training, transcribes

team communication and uses a custom natural language processing pipeline supported by GPT-4 to identify effective, incomplete and missed closed-loop communication instances. Comparable simulation-oriented designs also appear in HCI and vocational technical training: Barambones et al. (2024) use ChatGPT to support role-play interviews in HCI education, while Huang et al. (2025) develop a fine-tuned Llama 3-based automotive electronics internship assistance system, benchmarked against BERT-base and GPT-2. These studies use GenAI to transform static learning materials into interactive, scenario-based and feedback-rich training experiences.

In the third stream, LLMs are utilized as co-instructors, content designers, or knowledge-support assistants within structured educational and workforce training settings. Lu et al. (2024b) evaluate GenAI's capacity to support higher-order thinking in professional teacher development by co-generating pedagogical plans, reflective scaffolds, and hypothetical problem scenarios. In a different modality, Rezapour et al. (2024) explore how GenAI analyzes multiplayer game chat logs to identify implicit learning moments, thus facilitating evaluation of training progress and student engagement. Multi-modal integration also appears in video-based knowledge extraction, where models like GPT-4 are orchestrated alongside tools such as FFmpeg and Azure Cognitive Services. For example, de Almeida et al. (2025) present a pipeline in which learners upload instructional videos that are segmented and transcribed using FFmpeg and Azure APIs, after which GPT-4 evaluates the material, reinforces concepts and generates personalized quiz content. Extending this knowledge-support orientation to organizational training, Amaram et al. (2026) propose a multi-agent retrieval-augmented generation assistant for State Departments of Transportation, integrating retrieval, answer generation, evaluation and query refinement agents with a vision-language model, and validating the system on more than 500 technical documents and 100 queries.

5.1.3. HR information systems & analytics

Beyond the aforementioned HR functions, the use of LLMs to refine and standardize data streams represents an emerging area of research interest. Their advanced tokenization capabilities enable the generation of cleaned and denoised data frames, which underpin the improvement of the quality of downstream analytics. Within this context, agentic structures have proven effective, particularly in job-listing scenarios where data heterogeneity complicates integration with standard HR analytics systems (Chen et al., 2025b). Instead of treating GenAI merely as a conversational interface, studies in this stream position LLMs as upstream data-standardization layers that convert heterogeneous HR text into analytics-ready inputs. For instance, prompt-tuned ChatGPT agents automate text pre-processing, normalization and field extraction from job postings (Chen et al., 2025b); the LLM4Jobs framework converts HR documents into vector representations and maps them to canonical occupation IDs (Li et al., 2025b); and GPT-4o-based workflows extract contextual keywords, construct document-term matrices and feed structured skills databases that support dashboards on job demand, salary ranges and geographic trends (Chumwatana and Hpone, 2025). Across these studies, the main pattern is the movement from manual HR data preparation to LLM-mediated semantic standardization.

Within the broader effort to streamline data flows, HR analytics and decision support systems have emerged as key domains where LLMs offer transformative potential. Standardization layers, positioned atop downstream systems, enable real-time interaction with managerial personnel and support more informed decision-making. A central research focus involves linking employee performance data with organizational outcomes using LLMs. For example, Agarwal (2025) demonstrate how the natural language capabilities of GenAI can synthesize large performance datasets and clarify how individual outputs align with broader institutional goals. Building on this foundation, Nofal et al. (2025) propose a sentiment-based candidate scoring pipeline that uses LLMs to generate "hidden feedback", which is then processed through a RoBERTa sentiment classifier and a sigmoid function. This approach

is validated through robustness tests under different prompt variations and fairness assessments showing that changes in candidate demographics, such as country, religion and gender, can alter scores by up to 10%, depending on the underlying LLM.

5.1.4. Occupational health and safety

Recent studies show the potential of GenAI applications to serve as a unifying analytic layer that strengthens every stage of occupational health and safety (OHS) practice, from real-time risk assessment to personalized and scalable training. A common feature of this stream is the coupling of GenAI with sensing, computer vision, VR, RAG or multimedia tools, enabling systems to translate technical safety signals into actionable feedback for workers and managers. ERG-AI illustrates this capability by pairing computer vision posture classification with LLM narration: sensor analytics quantify worker posture and epistemic uncertainty, after which a pre-trained transformer, such as GPT-4 or LLaMA-7B, turns the raw metrics into plain-language risk reports and ergonomic advice (Sen et al., 2024). In a similar real-time advisory setting, Işıkdağ et al. (2025) fine-tune YOLOv8, YOLOv11 and YOLOv12 on a helmet-detection dataset and integrate the detection outputs with OpenAI GPT-3.5-turbo through a Streamlit-based interface, enabling immediate safety recommendations for construction-site managers. Generative chatbots further extend OHS support. The “Genie-on-Demand” GPT-3.5-turbo bot, fine-tuned on course PDFs, provides curriculum-specific answers that support hazard-identification skills, knowledge retention and confidence (Hakim et al., 2024), while iSafeTrainer embeds a GPT-4 dialog agent inside a Unity-based VR environment to tailor scenario difficulty, perform real-time PPE checks and provide corrective feedback (Hussain et al., 2024). Beyond applications implemented at the intersection of LLM and VR technologies, Chen et al. (2026) introduce ConSTRAG, a personalized construction safety training framework combining LLM-based agents, retrieval-augmented generation, knowledge graph reasoning, accident-report databases and a prototype mobile application, with benchmarking conducted through an 11,020-question construction management question bank and expert evaluation. Finally, de Almeida et al. (2025) combine GPT-4, Azure TTS and FFmpeg to transform incident reports into short narrated simulation clips for toolbox talks and daily briefings. Together, these applications show how LLM-powered systems can translate complex OHS data and domain knowledge into clear, context-aware feedback.

5.1.5. Workforce planning

Workforce planning is another HR function that could be powered by GenAI pipelines that convert unstructured labor market text and internal competence data into real-time intelligence on skill demand. Across this stream, GenAI is used to connect external labor-market sensing with internal competence planning. A prompt-tuned ChatGPT model structures 58k healthcare vacancies into titles, responsibilities and competencies, enabling planners to map regional shortages and highlight emerging roles such as “digital health strategist” (Chen et al., 2025b). In the IT industry, GPT-4o workflows extract skills from thousands of postings, cluster synonyms and rank terms with TF-IDF, producing dashboards that flag urgent gaps, salary bands and geographic hotspots (Chumwatana and Hpone, 2025). LLM4Jobs scales this approach by summarizing job descriptions and resumes, embedding them alongside ESCO/ISCO definitions and automatically assigning occupation codes, yielding clean and taxonomy-aligned datasets for long-range forecasts (Li et al., 2025b). Moving from external labor-market intelligence to internal competence planning, Barba et al. (2025) propose two LLM-enabled solutions for competence-based HRM in the aerospace industry: semantic taxonomy enrichment and role-competence embedding-based matching. Similarly, Akundi et al. (2025) use LLMs and AI agents to extract model-based systems engineering workforce competencies from LinkedIn job postings into structured skill, tool, education and experience fields, supporting curriculum alignment and sector-specific workforce planning.

Extending this internal planning orientation towards work-life balance and productivity analytics, Swamy and Adarsh (2025) introduce GenWorkBalanceNet, a hybrid GenAI-Deep Learning framework for predicting workforce productivity under alternative Work-Life Balance (WLB) policies in IT workforces. The framework combines a transformer-based encoder-decoder module that generates synthetic policy-driven scenarios, such as flexible schedules, hybrid work arrangements and wellness incentives, with an attention-enhanced LSTM and SHAP-based layer for forecasting productivity trends and identifying influential determinants. In this sense, it extends GenAI-enabled workforce planning beyond skill-demand forecasting towards counterfactual policy simulation. More broadly, the structured outputs produced by this stream, including competencies, urgency scores, occupation codes and policy-simulation insights, can inform recruitment, training, retention and productivity management.

5.1.6. Miscellaneous

The miscellaneous class comprises innovative GenAI applications that extend across several core HRM functions, including performance management, workforce planning, and retention analytics. These studies demonstrate how GenAI models such as GPT-4 and ChatGPT are being embedded into diverse HR processes: from enhancing the objectivity and scalability of employee evaluations, to enabling real-time labor market intelligence, competence mapping and job matching, and to forecasting attrition risk through integrated, multi-criteria decision frameworks.

Recent research positions GenAI as a double-edged sword for performance management, promising richer, data-driven assessments while simultaneously increasing the risk of embedded bias. On the positive side, GPT-4 exhibits high accuracy in judging soft skills such as communication, empathy, and ethics, even surpassing human assessors in controlled United States Medical Licensing Examination (USMLE)-style evaluations, signaling clear potential for scalable assessment and training tools (Brin et al., 2023). In a more directly managerial setting, Koppel et al. (2026) develop an OpenAI-based GenAI tool that processes employee self-reflection texts and structured performance metrics to generate meeting preparation summaries and discussion topics for quarterly 1:1 interviews, with human-professional evaluation showing that GPT-4o outperforms GPT-4o-mini across most assessment variables while substantially reducing preparation time. However, this promise is tempered by evidence that ChatGPT-3.5, when drafting promotion letters, produces systematically different tone, length and language depending on the sex of the candidate, reproducing rather than eradicating bias in high-stakes decisions (Desai et al., 2024). A broader assessment adds further nuance, showing that organizations are experimenting with GenAI chatbots for personalized on-boarding, transparent explanations of turnover risk, adaptive learning pathways and the linkage of granular performance data to organizational outcomes (Agarwal, 2025). These studies indicate that performance-management applications combine promising assessment-support capabilities with unresolved concerns regarding bias, transparency and overreliance on automated judgment.

GenAI is also redefining retention analytics by combining multi-criteria scoring with transparent, actionable insights. Abu-Faty et al. (2025) embed ChatGPT-4 as a “virtual expert panel” that creates HR personas, elicits and weights 15 turnover drivers through AHP pairwise comparisons, rates employees on the Saaty scale and passes the resulting matrices into a TOPSIS engine, both ranking exit risk and producing a synthetic expert-labeled dataset for downstream churn classifiers. Complementary work shows how predictive engines can be paired with explainable GenAI dashboards, where LLMs reveal the rationale behind exit-risk scores and chatbots provide personalized on-boarding and stress-management guidance to prevent early attrition (Agarwal, 2025). Similarly, Wyawahare et al. (2025) combine Feedforward Neural Network-based attrition prediction with an LLM-powered AI Chat

Table 2
Generative LLM applications across HRM functions, organized by theme.

HRM function	Objective	Technical task	Implementation pathways (decoder-inclusive schemes)	
			In-house	API-based
Recruitment & Selection	Resume screening & Candidate ranking	Multi-stage resume filtering, summarization, grading and shortlisting	LLaMA 2	GPT-3.5; GPT-4
		Synthetic resume generation, resume completion and candidate–job matching Adaptive/generative job recommendation with preference alignment and bias mitigation	Fine-tuned LLMs GIRL; reward model; PPO	GPT-3.5; GPT-4 GPT-3.5-turbo
	Interview assessment & Candidate evaluation	Virtual technical-interviewer scoring Medical/residency selection benchmarking and application scoring Personality, leadership and competency assessment from video or self-interview responses Candidate simulation and adversarial response generation for selection tests	Mistral 7B LLaMA 2 Qwen 32B LLaMA 2	GPT-4; Gemini Bing Chat; Bard; ChatGPT-3.5; GPT-4 GPT-3.5; GPT-4; GPT-4o-mini GPT-3.5; GPT-4; Jasper
	Job-Ad generation	Vacancy parsing, keyword optimization, skills-dictionary construction and JD structuring	Mixtral-8 × 7B	GPT; GPT-4; GPT-4o
	On-boarding support	Chatbot-driven personalized guidance, stress-management and MCDA-supported role optimization		GPT-4
Training & Development	Personalized career coaching	CV analysis, tailored interview preparation, job-alignment support and feedback loops		GPT-4
	Simulation & Experiential learning	Healthcare, nursing and ACLS simulation with multimodal feedback and communication assessment VR/metaverse interview practice and HCI role-play learning Vocational technical training and domain-specific Q&A support	Custom NLP pipeline Fine-tuned Llama 3	GPT; GPT-4; DALL-E 3 GPT
		Lesson plans, reflective scaffolds and hypothetical problem scenarios Learning analytics from game chats and video-based quiz/content generation Multi-agent RAG for technical knowledge management and workforce training	RAG agents; VLM	GPT GPT-4 LLM
	AI content Gen. & Knowledge support			
HR information systems & Analytics	Data standardization & Pre-processing	Job-listing tokenization, normalization, field extraction and HR-document vectorization	Mixtral-8 × 7B	GPT-4-turbo
	Skills & Labor-Market analytics	Keyword extraction, skills databases, document–term matrices and dashboard generation		GPT-4o
	Performance analytics & Decision support	KPI/performance-data synthesis for organizational decision support Sentiment-based candidate scoring and bias detection through prompt variation	RoBERTa classifier	GPT-4 GPT
Occupational health & Safety	Risk Assessment & Reporting	Posture/helmet analytics translated into ergonomic or construction-site safety recommendations	LLaMA 7B; YOLO models	GPT-3.5-turbo; GPT-4
	Interactive safety training	Curriculum-tuned Q&A chatbot for hazard-identification learning VR safety trainer with adaptive scenarios, PPE checks and corrective dialog Personalized construction safety training with RAG, knowledge graphs and worker-specific assessment content	RAG agents; knowledge graph	GPT-3.5-turbo GPT-4 LLM
	Safety content generation	Incident-report transformation into narrated simulation videos		GPT-4
Performance management	Data-Driven evaluation	Soft-skill scoring, chatbot feedback loops and performance-data interpretation 1:1 meeting preparation from self-reflections and structured performance metrics		GPT-4 GPT-4o; GPT-4o-mini
	Bias analysis & Promotion narratives	Promotion-letter gender-bias audit and LLM-generated evaluation narratives		GPT-3.5
Workforce planning	Labor-Market intelligence	Vacancy parsing, skills extraction, shortage mapping and hot-spot dashboards		GPT; GPT-4o
	Occupation coding	JD/resumé summarization and mapping to ESCO/ISCO occupation codes	Mixtral-8 × 7B	
	Competence mapping & Strategic alignment	Taxonomy enrichment, role–competence matching and MBSE competency extraction	LLMs; AI agents	
	WLB policy simulation & Productivity forecasting	Synthetic WLB-policy scenarios and temporal productivity prediction	Transformer encoder–decoder; LSTM	
Retention /Attrition	Turnover-Risk forecasting	AHP + TOPSIS with virtual expert panels and synthetic expert-labeled churn data Explainable dashboards and stress-management chatbots for proactive retention Neural attrition prediction with LLM-based risk explanation and retention recommendations	Feedforward Neural Network	GPT-4 GPT-4 LLM-powered AI Chat Widget

Widget that explains risk factors and generates retention recommendations for HR decision support. These approaches showcase how GenAI can quantify churn likelihood while providing managers with interpretable levers for proactive retention.

To summarize the main findings within the GenAI application class, we refer to [Table 2](#), which consolidates the main literature streams, HRM-specific objectives, technical tasks and corresponding implementation pathways designed to meet the identified HRM needs. Following the preceding analysis, the table should be read as a synthesis layer illustrating the current state of generative, LLM-centric applications in HRM. At the objective level, API-based pathways dominate most HRM functions. For example, they constitute the main implementation pathway for almost all methodologies developed to support tasks related to Recruitment & Selection and Training & Development, with most approaches relying on GPT-centric implementations. In HR Information Systems & Analytics and Occupational Health & Safety, some in-house or fine-tuned/open-source schemes also emerge, although more selectively, especially in HR-document vectorization, occupation-code mapping, posture/helmet analytics and RAG-supported safety training. Overall, platform-mediated GPT-family solutions emerge as the most frequently adopted implementation scheme, although several studies increasingly combine them with in-house models, classifiers, RAG agents or domain-specific fine-tuning. To facilitate a clearer understanding of the elements presented, we have adopted a consistent nomenclature for GPT models, following the format Model Name – Generation (e.g., GPT-4o), wherever this information was explicitly provided in the reviewed sources. In cases where the original articles did not specify the model generation, we have referred to the model using only its generic name (e.g., GPT), without inferring the underlying version.

5.2. Discriminative usage of LLMs

5.2.1. Recruitment and selection

Within the broader spectrum of HRM, discriminative applications of LLMs have gained notable traction in recruitment-related tasks. Unlike generative tasks that produce novel text (e.g., job ads or feedback reports), discriminative LLMs focus on evaluating, classifying, ranking, or matching inputs based on learned semantic representations. These models operate by transforming unstructured textual content, such as resumes, job descriptions, or application records, into high-dimensional embeddings that can be used for predictive or comparative tasks. Their use is particularly suited to solve long-standing recruitment challenges, including resume selection, candidate–job fit estimation and applicant ranking, thus automating early-stage selection and candidate-screening processes.

The first and most widely explored stream of LLM-centric research in recruitment involves the discriminative use of language models for semantic matching and decision support. This research primarily aims to automate early-stage processes such as resume screening, candidate shortlisting and person–job fit evaluation, using transformer-based embeddings and similarity-based ranking techniques. Within automated resume screening, several studies combine LLM-derived embeddings with classical or deep classifiers. For example, [Tian et al. \(2023\)](#) introduce a hybrid architecture that integrates BERT embeddings with Latent Semantic Analysis (LSA) topic vectors and Support Vector Machine (SVM) classification to improve resume-based filtering. Similarly, [Deshmukh and Raut \(2025\)](#) develop a keyword-enhanced similarity pipeline that uses BERT-generated vectors and cosine similarity to rank candidates against job descriptions. More integrated frameworks, such as that of [Sareddy and Farhan \(2025\)](#), utilize a domain-adapted BERT variant (Nish-BERT) with a Gated Attention-on-Attention Recurrent Unit (GAARU) to evaluate resume–job compatibility, while embedding recruitment transactions within a secure blockchain infrastructure.

In a similar resume-processing direction, [Peng \(2025\)](#) propose a three-stage LLM-based framework for automated resume screening,

where fine-tuned LLaMA2 performs sentence-level classification of resume components, supports merit-based grading and summarization through an HR-agent paradigm, and feeds either automated or HITL candidate-selection decisions. Extending this LLaMA-based orientation, [Deshmukh et al. \(2026\)](#) develop a scalable resume screening system based on LLaMA 3, where resumes are converted from PDF into structured text, combined with the job description and processed to produce candidate–job relevance assessment, ranked candidate lists and reasoned explanations for selected applicants. Although such systems may include explanatory text, their core utility remains discriminative, since they evaluate resume–job fit, rank candidates and automate initial screening. Sentence-BERT has also been used for contextual similarity scoring and automated shortlisting, while [Werner and Laber \(2024\)](#) apply BERTimbau to extract structured information from Portuguese resumes through sequence labeling. On a broader scale, [Naudé et al. \(2023\)](#) present AIRM, an AI recruiting system that combines RoBERTa-based sentence embeddings, Facebook AI Similarity Search (FAISS)-based approximate nearest-neighbor retrieval and Cross-Encoder re-ranking to support candidate screening and ranking in the Saudi labor market. Similarly, [Bocharova and Malakhov \(2024\)](#) fine-tune a BERT-based model, namely “SkillPhrase2Vec”, with special boundary tokens and contrastive learning, so that skill phrases in resumes or vacancies receive context-aware embeddings for candidate-skill identification, normalization and matching.

Complementing these text-parsing pipelines, other approaches build on hybrid schemes that incorporate visual document understanding, OCR-based extraction, semantic matching and generative LLM modules into unified recruitment workflows. For instance, [Zahour et al. \(2026\)](#) introduce an automated resume parsing and ranking system that combines YOLOv8-based visual segmentation, Tesseract OCR and OpenAI/Gemini LLM APIs to extract resume sections, compute job–resume similarity scores and rank candidates according to predefined recruitment criteria. Beyond this multimodal parsing orientation, [Hepzibah et al. \(2025\)](#) propose a hybrid resume screening pipeline in which Sentence-BERT embeddings and cosine similarity perform candidate–job matching and threshold-based shortlisting, while an LLM module generates concise candidate summaries and supports explainable recruiter review through an HR dashboard. Similarly, [Ifakir et al. \(2025\)](#) combine NER-based resume parsing, RoBERTa/CamemBERT-driven semantic matching and cosine similarity scoring with a generative LLM component that produces entity-based interview questions after a candidate–job match is confirmed. In a related architecture, [Chong et al. \(2026\)](#) introduce an integrated AI recruitment system that combines a Resume Screening AI and a Job Recruitment Chatbot, where generative extraction and conversational interaction are coupled with discriminative resume–job comparison, relevance scoring, classified evaluations and Q&A validation through TF-IDF and fine-tuned BERT. These contributions can therefore be positioned as hybrid LLM-enabled recruitment pipelines, in which generative extraction, summarization or candidate interaction support discriminative screening, validation and ranking functions.

A second stream, closely connected to selection but positioned at the interview stage, concerns hybrid LLM-based systems for interview orchestration and candidate assessment. In this direction, [Tigani \(2025\)](#) propose a parametric AI copilot for HR interviews, built upon a locally deployed Qwen3 LLM, a pretrained sentiment analysis model and real-time KPI computation. The implementation is hybrid by design: its generative module dynamically produces customized interview questions according to job-specific criteria and question difficulty, while its discriminative module evaluates candidate responses, analyzes emotional tone, scores hard, soft and language skills, and ranks candidates through a weighted global index. The copilot is framed as a human–machine collaboration mechanism, since recruiters retain control over the final decision while the AI agent standardizes question generation, response evaluation and real-time decision support during the interview process.

A third, more specialized sub-stream focuses on automated person-job fit and job matching, moving beyond surface-level screening to model deeper alignment between candidates and job requirements. These studies apply pre-trained transformer-based language models, such as BERT, T5 and Pegasus, to generate semantic embeddings of resumes and job descriptions for classification or recommendation tasks. In El-Deeb et al. (2025), pre-trained summarization models (BART, T5, BERT, Pegasus) condense verbose job descriptions into semantically dense inputs, which are subsequently vectorized using TF-IDF to support cosine similarity-based recommendation. Meanwhile, Rosenberger et al. (2025) introduce a fine-tuned BERT-based Siamese network that embeds resumes and standardized ESCO job descriptions in a shared vector space, enabling cross-domain job recommendation validated through human expert feedback. Moving towards personality-informed matching, Alqudah et al. (2025) apply transfer learning with BERT, RoBERTa and XLM-RoBERTa to classify soft skills and Big Five personality traits and map them to functional areas, supporting candidate-job recommendations beyond hard-skill matching.

Beyond these approaches, advanced end-to-end architectural paradigms for classification have also been proposed. In Xue et al. (2025), local contextual embeddings from BERT are fused with global resume features extracted from a knowledge graph using a Kolmogorov–Arnold Network (KAN), achieving over 94% accuracy. In another graph-based direction, Frazzetto et al. (2025) formulate candidate-job matching as an inductive graph learning problem by extracting candidate and job attributes with LLMs, converting them into embedding-based bipartite graphs and comparing multiple Graph Neural Network architectures for binary and ordinal screening tasks on real-world recruitment processes. Finally, Wang et al. (2021) combine BERT-based sentence embeddings with TransE knowledge-graph vectors and feed them into a multilayer attention network to assess fit scores. These studies differ in architectural choices, from TF-IDF hybrids to graph neural networks and attention layers, but they share a focus on embedding resumes and vacancies in task-specific latent spaces for person-job matching.

Fraud detection in online recruitment advertisements is a specialized topic that can be situated within the broader domain of job-fitting approaches, as it screens out inauthentic or deceptive job listings before they distort candidate matching or compromise user trust. Recent research has leveraged discriminative LLM-based classifiers to address this challenge. For instance, Naudé et al. (2023) show how fine-tuned transformer models such as BERT, RoBERTa and XLNet can classify job postings as fraudulent or legitimate and also distinguish fraud subtypes, such as identity theft or multi-level marketing scams. Elaborating upon this direction, Taneja et al. (2025) introduce Fraud-BERT, a customized BERT variant that integrates contextual embedding and semantic extraction modules to capture subtle textual cues, such as vague job descriptions or unnatural phrasing. These studies position LLM-based classifiers as upstream filters that help secure e-recruitment platforms and maintain the integrity of downstream job-matching systems.

A final stream of recruitment literature resides at the intersection of selection and HR information and decision support systems, examining how discriminative LLM-driven techniques automate selection tasks while generating structured data for organizational decision-making. In Qostal et al. (2024), resume texts are transformed into BERT embeddings and fed into CNN-GRU classifiers, achieving 93.5% precision in specialty assignment and producing structured data for HR dashboards and employability analytics. The framework presented in Jain et al. (2024) extends this paradigm by combining BERT-derived text embeddings with CNN audio-video features to predict Big Five traits, which are then used to match candidates to roles and populate decision support modules for turnover forecasting. Similarly, Akram et al. (2024) fine-tune BERT and RoBERTa with SMOTE-balanced corpora to flag fraudulent postings and integrate fraud-check outputs into recruitment workflows and analytics pipelines. On the workforce-planning side, Guo et al. (2025) employ a BERT-BiLSTM-CRF pipeline for entity extraction

and LDA topic modeling to map market demand trends, feeding time-series forecasts into strategic HR dashboards. Relatedly, Podolchak et al. (2025) model job descriptions as a multi-label classification task using a multilingual BERT architecture to support automatic detection of working conditions and social benefits in vacancy texts.

Additional technical contributions further connect discriminative screening with HR analytics. Bevara et al. (2025) develop Resume2Vec, a system that generates dense vectors via multiple transformer models, such as GPT-4 and Gemini, to replace keyword matching with context-aware ranking inside ATS analytics. Wang and Zhu (2022) demonstrate how BERT-based named entity recognition and the BERT-BiLSTM-CRF approach can populate a recruitment-entity knowledge base, allowing HR analytics to query qualifications, skills and personality requirements at scale. Within personnel-selection decision support, Hoque et al. (2026) combine fine-tuned DistilRoBERTa-based multi-class classification of LinkedIn profile attributes with Fuzzy-TOPSIS ranking, translating transformer-derived proficiency scores into interpretable candidate rankings aligned with human expert evaluations. Collectively, these studies show how discriminative LLMs can support both operational selection and semantically structured HR decision analytics for talent acquisition and workforce planning.

5.2.2. Training and development

In the field of training and development, the discriminative application of LLMs is used to address issues related to personalization, scalability and domain-specific training. Compared with GenAI models, discriminative LLMs are primarily used for evaluation and prediction tasks, including ranking learning needs, assessing competency development and personalizing recommendations through learner inputs and semantic embeddings. These models can analyze unstructured data on learner engagement, task performance or pedagogical discussion to provide actionable information to trainers and organizational development planners.

In our analysis, we identified three subcategories in which discriminative LLMs are integrated in development and training environments: (a) career and coaching personalization, (b) simulation and competency attainment in domain-specific or soft skills, and (c) semantic inference of learning interactions for adapting instruction. In the first subcategory, LLM-based systems are used as individualized coaching agents that infer user goals, career aspirations and development paths. In virtual safety training, for example, Sabir et al. (2025) develop a ChatGPT-based virtual reality assistant that interacts with employees in real time. Despite leveraging a generative language model, the system relies on discriminative AI components, such as intent recognition and response classification via rule-based and ML classifiers, to interpret worker input, identify hazard-related intents and tailor safety coaching accordingly.

The second subcategory concerns domain-specific or soft motor skill assessment using interaction monitoring and semantic interpretation. Iranfar et al. (2025b) assess the performance of surgical trainees by integrating soft sensor data with NLP-based classification of spoken reflection. Their discriminative learning model detects psychomotor learning states, such as expert imitation versus procedural mastery, from brief narrative utterances to provide immediate feedback. Similarly, Xu and Luo (2022) use BERT-based models to classify reflection statements of student-teachers across dimensions of professional identity development, supporting adaptive scaffolding related to teaching competence maturation. The third stream deals with adaptive instructional inference based on discriminative interpretation of student-generated content. Rezapour et al. (2024) analyze chat data from multiplayer games using fine-tuned LLMs to estimate learning progression. The system labels user messages according to learning stages such as exploration, hypothesis and consolidation, enabling adaptive support in collaborative game-based learning sessions. In a related e-training analytics setting, Alotaibi et al. (2025) propose an enhanced RoBERTa model with domain-adaptive pretraining and dynamic attention scaling to analyze open-ended training feedback and classify employees into job-performance categories, thus supporting personalized interventions and data-driven training decisions.

5.2.3. HR information systems & analytics

Advancing recruitment decision-making through LLM-driven analytics has emerged as a significant research focus at the intersection of talent acquisition and information systems. Recent work treats transformer models not only as language engines but also as upstream standardization layers that generate structured, analytics-ready data from resumes, job ads and interview transcripts. Once integrated into HRM pipelines, these enriched representations underpin dashboards and decision support modules that inform HR functions such as staffing, development and workforce planning.

The first analytic strand centers on creating an enterprise “skills graph”. Boundary-aware BERT models such as SkillPhrase2Vec embed and cluster skill phrases, align them with public ontologies such as ESCO and Occupational Information Network (O*NET), and continuously expand corporate taxonomies; the same pipeline quantifies in-house competencies and highlights skill gaps, enabling targeted up-skilling initiatives and evidence-based recruitment (Bocharova and Malakhov, 2024). Complementing this, resume-level embeddings supply deep classifiers that sort candidates by engineering specialty with high accuracy levels (>90%) and feed employability dashboards that visualize student skill deficits against labor-market demand (Qostal et al., 2024). Transformer-based similarity scoring, as in Resume2Vec, replaces keyword filters inside applicant tracking systems, providing candidate rankings and real-time analytics on shortlist composition, diversity and pipeline velocity (Bevara et al., 2025). Such types of extraction outputs have also found to power named-entity pipelines that build recruitment-entity knowledge bases, automate taxonomy maintenance and enable advanced queries spanning qualifications, experience and personality trait assessments (Wang and Zhu, 2022).

In addition to these approaches, multi-modal architectures have been proposed to fuse LLM text embeddings with audiovisual cues to infer Big Five personality traits. The resulting feature vectors populate data warehouses and drive dashboards that forecast placement success, satisfaction and turnover risk, thereby informing role assignments and retention strategies (Jain et al., 2024). Since job-matching systems depend on data integrity, fraud detection also becomes relevant to HR analytics. In this direction, Taneja et al. (2025) introduce upstream transformer-based classifiers to flag fraudulent job advertisements and safeguard online recruiting processes. Finally, large-scale vacancy mining combines BERT-based entity extraction with topic modeling, co-occurrence graphs and time-series forecasting to surface macro-level skill demand, regional hot spots and emerging role profiles, equipping HR leaders with forward-looking intelligence for budget allocation, campus-hiring targets and re-skilling road maps (Guo et al., 2025).

Beyond job-matching analytics, LLMs have also been studied for multilingual text mining purposes that target job-satisfaction insights. In this vein, Lee et al. (2025) propose an analytics pipeline in which employee reviews are first embedded with Sentence-BERT and then analyzed using Contextualized Topic Modeling (CTM), with topic quality refined through normalized point-wise mutual information (NPMI). The pipeline uncovers HR-salient themes, such as salary, attendance and welfare, and visualizes their prevalence across countries and employment categories through interactive dashboards.

5.2.4. Miscellaneous

The miscellaneous category comprises discriminative LLM applications that extend across HRM functions not represented as standalone classes, primarily in the areas of occupational health & safety, workforce planning, performance management and retention analytics. These studies demonstrate how transformer-based and LLM-enhanced models can support HRM decision-making beyond recruitment and training, by classifying safety incidents, detecting psychosocial risks, extracting standardized labor-market competencies, transforming unstructured feedback into performance-relevant labels and extracting employee-experience themes that inform retention strategies.

Occupational health & safety constitutes a distinct HRM function within this class, as recent studies increasingly use transformer-based and LLM-enhanced models to support safety monitoring, risk identification and preventive decision-making. In this direction, Danish and Chatterjee (2026) address the lack of safety-specific language models by adapting BERT and ALBERT to occupational safety corpora through domain-adaptive pretraining, producing safetyBERT and safetyALBERT. These models are evaluated on accident type, equipment, injury degree, claim type and injured body part classification tasks across industrial safety datasets. In a complementary workplace-safety setting, Hasan et al. (2025) develop an occupational stress detection framework that combines machine learning, deep learning and LLM-based domain analysis, converting tabular workplace survey data into text and evaluating stress classification models to support early identification of safety-critical psychosocial risks. Focusing on chemical risk assessment, Pekel et al. (2026) implement a GPT-based pipeline for extracting fixed variables from safety data sheets, such as company name, product name, hazard statements and physical status, thereby automating the transfer of chemical-risk information into online inventories. Finally, Zhang et al. (2025) propose an LLM-driven framework for construction worker activity recognition, where helmet-mounted inertial measurement unit signals are transformed into semantic meta-event features and interpreted by an agent network to infer worker activity patterns relevant to safety monitoring. Extending this safety-monitoring orientation to multimodal chatbot-based inspection, de Marco et al. (2025) evaluate a low-code multimodal conversational agent for construction-site safety, where real-world site images are processed to identify hazards, recommend preventive measures and assess compliance with Italian Legislative Decree 81/08. Although the system also produces structured textual explanations and safety recommendations, its core utility is discriminative and evaluative, since the model is primarily used to detect, classify and score occupational-risk conditions from visual site evidence rather than to generate open-ended safety content.

Workforce planning is also supported by discriminative LLM applications that extract structured competency signals from unstructured labor-market text. In this direction, Kavargyris et al. (2025) introduce ESCOX, an open-source tool that combines LLMs, sentence-transformer embeddings, ESCO embeddings and cosine similarity to extract skills, skillsets and occupations from job postings and general text, mapping them to standardized ESCO and ISCO-08 categories. From an HRM perspective, this taxonomy-aligned extraction mechanism supports skills gap analysis, workforce intelligence, recruitment analytics, curriculum alignment and policy planning, while its technical contribution remains discriminative because the system identifies, classifies and ranks standardized skill and occupation categories rather than producing open-ended generated content.

Also in tasks relating to the performance management function, LLM-enhanced text classifiers have recently been recognized as promising alternatives towards feeding performance dashboards with objective skill ratings. In a surgical simulation study, Iranfar et al. (2025b) use an on-premise pipeline that couples decoder-only models, namely Claude 3 Opus, GPT-4 and Gemini Pro, with BERT, using rule-based prompts to map instructor comments to novice, intermediate or expert levels. These labels provide trainers with real-time insight and can be inserted into formal appraisal metrics. From an HRM perspective, the value of this approach lies in transforming qualitative instructor feedback into standardized performance indicators that can support more consistent skill assessment, appraisal and training-related performance monitoring.

Retention analytics also appears as an HRM function in which discriminative LLM-based text analytics can convert large-scale employee voice data into actionable organizational insights. For instance, Lee et al. (2025) feed Sentence-BERT embeddings into a Contextualized Topic Model (CTM) to analyze 128 000 Glassdoor and Jobplanet reviews. The model identifies HR themes, such as pay, attendance management and welfare support, and quantifies the impact of each theme

on job satisfaction ratings with a path-analysis coefficient T_{ij} . The resulting dashboards compare topic prevalence and sentiment across the United States and South Korea, as well as between current and former employees, supporting culture-sensitive retention analysis.

To summarize the main LLM-centric contributions, stemming from the field of discriminative AI applications, we refer to Table 3, which consolidates the main literature streams, HRM-specific objectives, technical tasks and corresponding architectural paradigms designed to deliver the identified HRM needs. Following the preceding analysis, the table should be read as a synthesis layer illustrating how discriminative, LLM-centric applications are mapped across HRM functions. At the objective level, the mapping covers 22 objective streams across seven HRM functions. Encoder-only architectures emerge as the dominant paradigm, appearing across all mapped HRM functions and supporting tasks such as resume–job description ranking, person–job matching, fraud detection, skill-taxonomy alignment, safety-risk classification and employee-review theme mining. Decoder-only schemes appear more selectively, mainly in hybrid pipelines for resume parsing, interview orchestration, safety-risk analysis and performance-feedback classification, while encoder–decoder models are primarily observed in person–job matching and forecasting-oriented analytics.

5.3. A context-specific mapping of LLM-centric tools in HRM

A critical assessment of the evidence presented within this section highlights multiple LLM-centric research streams implied in the field of HRM. Towards exemplifying the process of developing an LLM-centric application in the field under consideration, we consolidate the information presented in the previous subsections in order to develop an objective-driven map, which details the most prominent implementation pathways. To perform this task, we elaborate on the main argument of this work, which is the different LLM applications in generative and discriminative approaches.

These two different approaches serve as the backbone of the mapping, presented in Fig. 6. The corresponding diagram provides an executive mapping that aligns six horizontally ordered layers, thus translating the fragmented discussion of the adoption of LLMs in HRM into a single, end-to-end value chain. At the left-most level, high-level HRM functions define the strategic problem space. These functions cascade into concrete operational objectives such as on-boarding, personalized training, and gap analysis, which in turn connect to an AI-paradigm layer that distinguishes generative from discriminative approaches. Specific pre-trained models then instantiate each paradigm: GPT-4, Gemini, BART, GPT-3.5, and fine-tuned LLaMA-2 occupy the generative branch, while BERT, RoBERTa, Sentence-BERT, and XLNet anchor the discriminative branch. The fourth plane captures LLM-enabled HR tasks—chatbot on-boarding, curriculum personalization, semantic ranking, sentiment and KPI tracking, and so forth, demonstrating how each model class delivers the earlier objectives. Finally, implementation pathways that include prompt engineering, fine-tuning and HITL oversight expose the governance mechanisms required to translate model output into compliant organizational action. By following this structural approach, the diagram enables managers to pinpoint an HR pain-point, identify the paradigm and model families most validated for that context, and anticipate the requisite oversight levers, while simultaneously offering researchers a gap-analysis lens that highlights under-explored intersections of objectives, model classes, and governance schemes.

To facilitate a better understanding of this diagram, we note that only the HR functions attracting the greatest research interest are displayed. Specifically, the mapping emphasizes HRM functions with the strongest concentration of evidence in at least one of the two technical classes, namely generative or discriminative LLM applications. For this reason, workforce planning and performance management are retained, as they emerge as sufficiently coherent streams in the GenAI and/or discriminative parts of the reviewed sample, even though

related contributions may otherwise appear within broader residual categories. We also mention that the relative size of the GenAI and Discriminative AI blocks is indicative of their proportional presence in the application-oriented subset of the sample (i.e., approximately 52% of these rely on GenAI principles). Finally, the BERT-based hybrids node aggregates recurrent BERT-derived, domain-adapted, multilingual and pipeline-level variants, while BERT, RoBERTa and Sentence-BERT are retained separately due to their distinct use in the reviewed studies.

Beyond this mapping, some critical aspects reflecting cross-study patterns, contradictions and boundary conditions can be derived from the themes identified in the preceding analysis. Taking a critical stance, the evidence suggests that generative and discriminative LLMs should not be interpreted as two isolated application families. They operate as complementary utilities within broader HRM workflows. GenAI-oriented studies are mainly used when the task requires interaction, explanation or content production, as in on-boarding support, job-ad processing, interview-question generation and personalized training. However, several GenAI implementations also support structured extraction, profile enrichment, feedback preparation and decision-support communication. Discriminative applications are mainly used when the task requires classification, ranking, matching or scoring, as in resume screening, person–job fit estimation, fraud detection and skills extraction. Technically, this distinction is reflected in the dominant implementation patterns: GenAI studies usually rely on prompt engineering, API-based models, RAG agents or fine-tuned decoder/encoder-decoder models, whereas discriminative studies rely more often on embeddings, similarity scores, classifiers, cross-encoders and task-specific fine-tuning. In several recruitment workflows, these two paths are combined, with discriminative components structuring and ranking information, and generative components summarizing results, preparing explanations or supporting human interaction.

A second critical observation concerns the uneven maturity of LLM-centric applications across HRM functions. Recruitment and selection represent the most developed area, as several studies move beyond conceptual discussion towards prototype implementation, baseline comparison, human evaluation or platform-level testing. Training and development, HR analytics, occupational health and safety, and workforce planning show intermediate maturity. Their technical evidence is mainly based on simulation environments, RAG-supported systems, multimodal prototypes, semantic standardization pipelines and dashboards. By contrast, performance management and retention analytics remain more exploratory. These areas are technically promising, but they involve high-stakes employee outcomes and therefore require stronger evidence on fairness, interpretability, longitudinal stability and organizational impact. This maturity pattern cuts across the generative–discriminative distinction: discriminative systems appear more mature where labeled data and benchmarkable outputs exist, particularly in embedding-based, classifier-based and hybrid pipelines where the final HRM utility remains ranking, classification or scoring. GenAI systems progress more clearly where expert feedback can assess explanation quality, interaction quality, generated content or decision-support usefulness.

This uneven maturity is closely connected to robustness. Stronger evidence appears when studies test models against baselines, use more than one dataset, include expert assessment, examine prompt sensitivity, report fairness checks or compare outputs with human decisions. Robustness is weaker when applications rely on a single model, a small or synthetic dataset, a narrow occupational setting or metrics that do not reflect real organizational consequences. From a technical perspective, this means that LLM-centric HRM systems should be evaluated not only by accuracy-like scores, but also by stability across prompts, transferability across job roles, consistency across demographic groups and compatibility with HITL review. Thus, the main implication of the mapping is that the taxonomy should be read not only as a descriptive classification of LLM applications in HRM, but also as a maturity- and robustness-sensitive map. It indicates where LLM-centric HRM research is technically advancing, where evidence remains fragmented, and where additional validation is required before these systems can be considered reliable components of organizational decision-making.

Table 3

Discriminative LLM applications across HRM functions, organized by theme.

HRM function	Objective	Technical task	Architectural paradigms		
			Encoder-only	Decoder-only	Encoder–Decoder
Recruitment & Selection	Resume screening & Candidate ranking	Resume–JD ranking, parsing, section segmentation, OCR-supported extraction, scoring, shortlisting and dashboard-supported review	BERT; Sentence-BERT; RoBERTa; CamemBERT; BERTimbau; Nish-BERT	LLaMA 2; LLaMA 3; Gemini 1.5 Flash	
	Interview assessment & Decision support	AI interview copilots for question orchestration, response evaluation, sentiment analysis, KPI computation and candidate ranking	Sentiment classifiers	Qwen3	
	Person–Job matching & Alignment	Technical-skill, soft-skill and personality-informed matching through semantic embeddings, similarity scoring and recommendation pipelines	BERT Siamese; RoBERTa; XLM-RoBERTa		BART; T5; Pegasus
	Fit classification (End-to-End)	Graph-fused fit scoring, knowledge-graph enriched matching and inductive graph learning for binary or ordinal screening	Domain-adapted BERT; BERT + TranSE		
	Job-Ad fraud detection	Fine-tuned job-ad fraud classification and subtype detection for recruitment-platform integrity	BERT; RoBERTa; XLNet; Fraud-BERT		
	Recruitment decision analytics	NER, topic modeling, job-posting quality assessment, LinkedIn-profile classification and multi-criteria candidate ranking	BERT-LSTM; BERT-BiLSTM-CRF; multilingual BERT; DistilRoBERTa	GPT-4; Gemini	
Training & Development	Personalized career coaching & Alignment	VR safety coaching with intent recognition, response classification and dynamic guidance	Rule-based/ML classifiers	GPT	
	Simulation & Competency attainment	Surgical trainee assessment via psychomotor signals and spoken-reflection classification Teacher-reflection classification for professional identity development	BERT BERT	Claude 3 Opus; GPT-4; Gemini Pro	
	Adaptive instruction via semantic inference	Learning-stage detection, training-feedback classification and personalized intervention support	Fine-tuned BERT; RoBERTa		
HR information systems & Analytics	LLM-based Data standardization	Parsing resumes/job ads; entity recognition; structured representation; skill-taxonomy and ontology alignment	BERT; SkillPhrase2Vec; BERT-BiLSTM-CRF		
	Analytics-grade representation pipelines	Resume embeddings, skill profiling, candidate similarity scoring and ATS-integrated context-aware ranking	Resume2Vec; Sentence-BERT; RoBERTa	GPT-4; Gemini	
	Data integrity & Governance	Fraud detection, validation layers and recruitment-data quality control	Fraud-BERT; BERT; RoBERTa		
	Insight generation & Forecasting	Vacancy mining, skill-trend prediction, employee-review topic mining and sentiment dashboards	BERT + CTM; Sentence-BERT; multilingual BERT	GPT	Gemini Pro
Occupational health & Safety	Risk classification & Detection	Safety-incident classification, psychosocial stress detection and fixed-variable extraction from safety data sheets	safetyBERT; safetyALBERT; ML/DL classifiers	GPT-based pipeline; LLM analysis	
	Safety monitoring & Risk assessment	Worker-activity recognition and multimodal construction-site hazard identification with structured risk scoring	Sensor classifiers	LLM agent network; multimodal chatbot	
Workforce planning	Competency extraction & Taxonomy mapping	Extraction, classification and ranking of skills, skillsets and occupations from labor-market text	Sentence-transformers; ESCO embeddings	LLMs	
	Workforce intelligence & Policy support	Skills-gap analysis, curriculum alignment, recruitment analytics and workforce-planning intelligence	Sentence-transformers	LLMs	
Performance management	Skill-level classification	Mapping free-text feedback to Novice/Intermediate/Expert or performance-relevant categories	BERT	Claude 3 Opus; GPT-4; Gemini Pro	
	Evidence-based dashboards	Integration of standardized labels into real-time appraisal, training and performance-monitoring metrics	BERT		
Retention/Attrition	Job-satisfaction theme mining	Multilingual employee-review embeddings, CTM topic extraction and satisfaction-theme identification	Sentence-BERT		
	Culture-sensitive dashboards	Path-analysis links between employee-experience themes and satisfaction across countries and employment groups	Sentence-BERT	GPT	

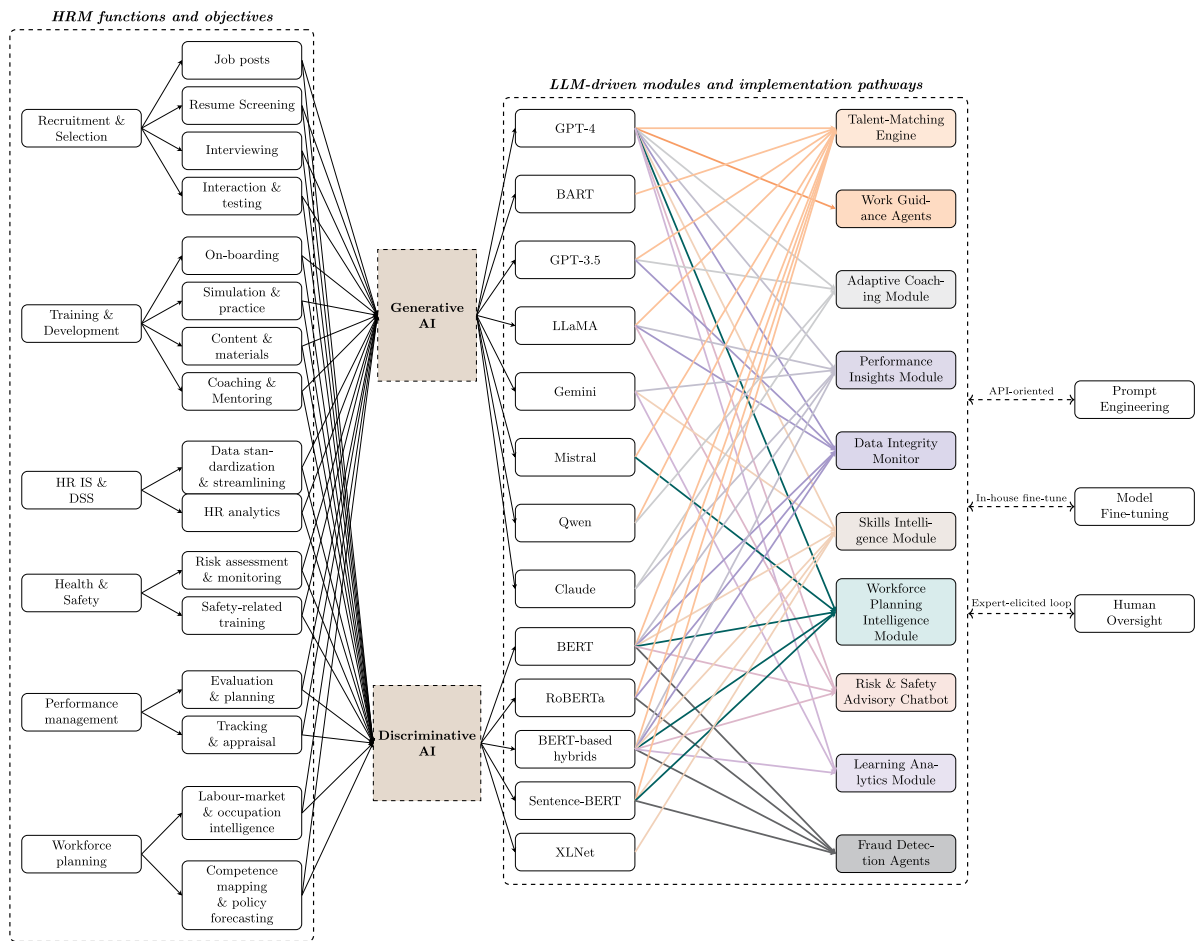


Fig. 6. A mapping of the common LLM-centric implementation schemes used to deliver HRM-specific objectives.

6. A critical synthesis and classification of theoretical contributions

To frame the review of recent scholarship on the integration of LLMs in HRM, we identified and coded 63 relevant articles using their stated methodological orientation as the basis for classification. In particular, this category includes articles of a theoretical nature, such as conceptual, empirical, ethical or prompt-focused studies. It is worth noting that within this classification, the interested reader will also find a subset of articles that incorporate statistical or empirical analysis approaches. These articles were retained within the theoretical group since their main objective was not empirical generalization, but rather the support or elaboration of new theoretical constructs, conceptual models or ethical frameworks based on sound statistical analysis.

6.1. Conceptual

The conceptual literature on GenAI and LLMs in HRM provides a multifaceted view of how these technologies are reconfiguring HR functions, raising ethical concerns, and reshaping individual skill development. The papers analyzed converge around three thematic axes: HRM augmentation, governance and theoretical framework and sustainable human development.

Several conceptual papers provide holistic frameworks, especially focusing on how LLMs can enhance core HRM functions through automation, data-driven personalization and predictive intelligence. For instance, in [Kliestik et al. \(2024\)](#) the authors describe a comprehensive integration of GenAI into the entire HR pipeline by exploring algorithmic hiring, VR-enabled training, performance dashboards and

turnover risk modeling where LLMs are portrayed as embedded cognitive engines that support both strategic and operational decision making. The study by [Langezaal et al. \(2025\)](#) examines how LLMs could be adapted for automated occupational coding (AOC), job-exposure matrices and occupational health assessment. The authors position LLMs as a promising but still unexplored extension of existing AOC systems and further argue that contextual language representations, fine-tuning and prompting may improve classification accuracy and generalizability. Similarly, [Chowdhury et al. \(2024\)](#) introduce a strategic HRM framework that uses GenAI to align individual development, task assignment and performance tracking with institutional goals. The authors draw on institutional entrepreneurial theory to argue that organizations must adapt HR practices to retain legitimacy in AI-saturated environments. Focusing on a public sector context, [Mwita and Kitole \(2025\)](#) highlight the role of GenAI in automating candidate screening, on-boarding, personalized training, feedback generation and strategic workforce forecasting. The paper draws on survey data from HR professionals and provides practical examples of how ChatGPT-style tools improve both efficiency and employee experience.

The growing influence of LLMs also raises pressing concerns about bias, transparency and the underlying limits of automation. In [Lievens and Dunlop \(2025\)](#) the authors propose a novel perspective by shifting the attention from organizations to applicants further highlighting how job candidates are increasingly using ChatGPT to optimize application materials, prepare for interviews and even manipulate personality and situational judgment tests. This dual-use scenario reveals a growing asymmetry between applicant capabilities and organizational preparedness. In a closely related contribution, the authors in [Campion \(2026\)](#) evaluate whether ML and AI systems can satisfy scientific, professional

and legal principles of personnel selection. The authors treat LLMs less as a fully evaluated HRM intervention and more as an emerging challenge for the validity and governance of AI-enabled selection. In [Korzynski et al. \(2023\)](#), the authors reflect on the epistemological shift prompted by LLMs in HR and beyond. They argue that GenAI transforms the conditions under which decision-making, communication and knowledge management occur, requiring an update to classical management theories. In HR-specific applications, the paper discusses how ChatGPT supports resume parsing, on-boarding communication and real-time employee support, re-framing traditional HR tasks within a continuously responsive digital infrastructure. Further developing the theoretical implications of GenAI for HRM, [Call et al. \(2026\)](#) examines how GenAI transforms value creation and value capture in knowledge work through the proposed “AI-specific Matthew Effect”. The paper challenges the view of AI as a simple workplace equalizer, arguing that domain expertise, early experimentation, metacognition and status-based attribution may allow star employees to capture disproportionate performance and compensation gains.

A third group of papers focuses on how GenAI impacts individuals’ careers, learning journeys, and long-term employability. [Tan \(2024\)](#) presents a compelling use case of a ChatGPT-based digital mentor for early-career researchers (ECRs). This system analyzes CVs and institutional goals to generate personalized advice on teaching, research and service development. It uses scoring mechanisms against the Researcher Development Framework and academic promotion criteria to offer gap-focused recommendations. Extending this to the broader workforce, the authors in [Adiasto \(2024\)](#) argue that effective interaction with LLMs is becoming a core employability skill. The paper outlines three specific training goals: prompt engineering literacy, AI-opportunity awareness (“generative-AI radar”) and quality control of AI outputs. Complementing this focus on sustainable employability, [Ruffin et al. \(2026\)](#) shows how LLMs can be used to audit whether online course content aligns with the skills required for twenty-first-century occupations. By linking MIT OpenCourseWare content to occupational skill demands, the study demonstrates that LLMs can scale the analysis of talent-development ecosystems across job families and bright-outlook occupations. In [El-mohandes and Marghany \(2024\)](#) the authors study ChatGPT’s utility in the hospitality industry’s recruitment processes where tasks are repetitive, multilingual and time-sensitive. The study reveals that ChatGPT can screen candidates, generate job ads, answer applicant questions, conduct virtual interviews, and support on-boarding.

6.2. Ethics/regulatory

Several authors have begun to examine the ethical and regulatory dimensions of GenAI integration into HRM. Specific conceptual/theoretical frameworks exist across a range of HR functions highlighting both opportunities and obligations in the evolving regulatory landscape of AI-driven workforce management. For example, the authors in [Andrieux et al. \(2024\)](#) develop a comprehensive ethical analysis of GenAI-enabled HRM by proposing the “Two-Rule Method”—“Do no harm” and “Do good” as their main guiding principles for evaluating the deployment of LLMs in organizational settings. In particular, the authors detail how LLMs could automate and enhance core HRM activities such as job-description maintenance, candidate screening, personalized training, benefits optimization and real-time performance feedback, while simultaneously highlighting risks around bias, oversight and fairness. The authors in [Baldassarre and Padovan \(2024\)](#) focus specifically on occupational medicine and workplace safety, arguing that LLMs can support health and compliance professionals by drafting regulatory communications, managing health-surveillance data and responding to safety inquiries. In their approach, LLMs are positioned as tools for improving regulatory adherence and institutional accountability, particularly through their integration into people analytics and HR information systems. Finally, the authors in [Abdelhalim et al. \(2024\)](#) propose a comprehensive set of safeguards for DEI in AI-enabled HR

systems. For instance, the authors emphasize that chatbots and generative models must be designed to prevent bias amplification and further support equitable outcomes in recruitment, training and employee mental health, among others. The authors discuss tools like Woebot and Wysa as concrete examples of AI-driven emotional support, thus reminding readers that well-being and fairness must be an integral part to technological design.

6.3. Prompt engineering

Prompt engineering has emerged as a foundational skill for both strategic HRM application and workforce development, especially given the ongoing incorporation of GenAI tools such as ChatGPT into organizational workflows. The authors in [Aguinis et al. \(2024\)](#) highlight how context-rich and well-structured prompts can transform LLMs into virtual HR advisors capable of delivering customized recommendations across a broad spectrum of HR functions like recruitment planning, compensation structuring, performance evaluation and strategic workforce design. The authors, through the use of detailed, scenario-based prompts, demonstrate that ChatGPT can generate operationally viable outputs such as KPIs, benefits roll-out plans and training blueprints, while also proposing eight prompt design guidelines and a verification framework to enhance reliability and ethical use. The authors in [Fed-eriakin et al. \(2024\)](#) frame prompt engineering itself as a critical 21st-century competency and they further conceptualize four dimensions (prompt literacy, structure, method and critical reasoning) as essential to enable effective human–AI collaboration. LLMs are positioned not only as tools, but as pedagogical environments in which users can iteratively develop the ability to formulate precise inquiries, evaluate AI outputs and apply them responsibly in professional contexts.

6.4. Statistical/empirical

Empirical studies that examine the application of LLM in HRM highlight a multifaceted transformation in core HR functions, particularly in recruitment and selection, training and development and performance evaluation. These studies often emphasize the dual goals of improving efficiency and fairness, while raising new ethical and methodological concerns about validity, bias and organizational readiness. The empirical literature reveals that LLMs are not merely tools for automation but are emerging as active cognitive agents in HRM processes. The research agenda now shifts from proof-of-concept validation to critical questions of trust, fairness, transparency and human–AI collaboration. Although performance gains and scalability are evident, the field must contend with new forms of ethical risk, especially with respect to bias amplification, cheating and algorithmic opacity.

Recruitment and selection emerges as the dominant theme within the statistical/empirical classification. For example, [Canagasuriam and Lukacik \(2025\)](#) shows that candidates who used ChatGPT for interview preparation received higher ratings, though at the cost of perceived honesty, raising concerns about integrity in asynchronous video interviews. Similarly, [An et al. \(2025\)](#) reveals algorithmic biases in resume evaluation, quantifying intersectional discrimination by feeding more than 360,000 synthetic resumes to multiple LLMs. The authors in [Gallo et al. \(2024\)](#) propose integrating dimensions of social perception (warmness and competence) into LLM audit frameworks to promote equitable hiring outcomes. Recent statistical and empirical studies show that LLMs are increasingly being tested and applied across many recruitment and selection activities, including recommendation-letter generation, applicant assessment, resume screening, bias auditing, job-posting analysis and recruitment-process automation. Other studies have examined whether AI-generated recommendation letters can be distinguished from human-written ones, showing that expert reviewers often struggle to reliably identify chatbot-generated content ([Simister et al., 2025](#)). Applicants’ use of GenAI during pre-hire assessments is also another theme. Relevant studies have found that

warning strategies, user motivations and perceptions of effectiveness strongly influence unauthorized GenAI use in hiring tests (Robie et al., 2026). A growing stream of work also investigates fairness and bias in AI-supported recruitment. Experimental simulations using GPT-4 demonstrated that gender cues in resumes and job descriptions can influence candidate evaluations and hiring justifications (Sivakamianathan and Musi, 2026), while large-scale audits across multiple professions showed that LLMs may reproduce biases related to gender, candidate ordering, and demographic indicators during candidate selection (Rozado, 2026). Similarly, simulated employer evaluations with ChatGPT revealed that demographic characteristics such as age, gender, sexuality and national origin can affect AI-generated hiring scores across different job types and model settings (Bilon, 2025). At the same time, LLMs are being used operationally to analyze recruitment-market data and improve HR analytics. Researchers applied the Qwen-2.5-14B model to classify long-term-oriented talent requirements in millions of Chinese job advertisements through prompt-engineered large-scale annotation workflows (Huang and Hu, 2026), while other work used ChatGPT Batch API to evaluate diversity, equity and inclusion signals in more than five million job postings from the perspective of minority applicants (Choi et al., 2026). Studies also empirically evaluated instruction-tuned foundation models for candidate screening, testing zero-shot and few-shot hiring predictions and explainability techniques for assessing candidate suitability (Pendyala et al., 2025). In organizational settings, case-study evidence suggests that ChatGPT can support several recruitment tasks, including job-ad drafting, interview-question preparation, candidate-document screening and HR communication generation, reducing recruitment time and operational costs (Koteczki et al., 2025). Furthermore, researchers proposed advanced resume-screening systems based on LLM embeddings combined with adversarial debiasing and contrastive learning to reduce gender and racial bias while maintaining matching performance between applicants and jobs (Wu et al., 2026). Finally, studies also highlight several emerging recruitment-related cybersecurity risks, showing that LLMs can assist in detecting covert malicious code embedded in technical hiring assignments sent to candidates during fake recruitment processes (Sanguino and de, 2026).

Recent empirical studies also show that LLMs are beginning to influence performance management and employee evaluation in various ways. For example, multimodal LLMs can support employee behavior monitoring in smart offices by specifically classifying workplace behaviors, generating structured HR reports and linking observed behaviors to performance evaluation (Chen et al., 2025a). GenAI use at work appears to improve innovative job performance through knowledge transfer, resource acquisition and job satisfaction, suggesting that employees benefit when AI helps them learn, access resources and collaborate more effectively (Zhang et al., 2026). Frequent GenAI use can also enhance adaptive performance, especially when employees perceive human–AI collaboration as less challenging and when organizations reduce obstructive stressors through training, clear guidance and careful monitoring of tensions (Yin and Wang, 2026). Finally, Tekin et al. (2025) and Brin et al. (2023) provide robust evidence that ChatGPT-4 and other models can assess soft skills and clinical performance with objectivity and inter-rater reliability comparable to, or even surpassing, human evaluators.

Recent empirical and prototype-based studies show that LLMs are increasingly being explored for strategic workforce planning, especially for skill matching, internal mobility, and career path prediction. One stream of research focuses on AI-enabled workforce restructuring through intelligent skill inference and credential validation. For example, SkillChain DX combines sentence-transformer embeddings, cosine similarity matching, blockchain credential verification, and HR policy dashboards to support dynamic role allocation, skill-gap analysis, and internal mobility decisions within public-sector HRM contexts (Al-Jaziri et al., 2026). Another line of work investigates predictive workforce planning through career path prediction models.

Comparative studies using fine-tuned LLMs such as Meta-Llama-3-8B-Instruct demonstrate that grounded LLM approaches can generate realistic next-role recommendations by combining structured career-history prompts with embedding-based occupation matching using ESCO labels and transformer models such as all-mpnet-base-v2 (Senger et al., 2025). Together, these studies suggest that LLM-based systems can support more adaptive and data-driven workforce planning by improving talent redeployment, career forecasting, and organizational skill alignment.

LLMs are also applied on occupational health and safety tasks in HRM. Current studies mainly evaluate whether LLMs can support risk assessment, occupational training, accident analysis, and employee safety decision-making. Several works demonstrate that models such as GPT-3.5, GPT-4 and GPT-4o can perform comparably to human experts in specialized safety contexts. For example, GPT models were tested on official Spanish occupational risk prevention examinations to evaluate their safety knowledge and reasoning abilities (García-Rudolph et al., 2025). Other empirical studies examine how employees use GenAI at work, further showing that both exploratory and exploitative GenAI usage can improve innovation and coping with workplace stressors, especially when employees perceive AI feedback as high quality (Yu et al., 2025). In applied safety analytics, LLM-based NLP workflows have been used to classify occupational accident mechanisms and job categories from unstructured text with hierarchical prompting and human validation procedures (Dunstan et al., 2025). Experimental comparisons also show that ChatGPT can generate construction safety risk assessments close to those of experienced occupational safety experts using standardized risk evaluation methods (Oral et al., 2025). Other studies demonstrate how contextualized LLMs can provide personalized exposure assessment, hazard identification and mitigation recommendations for workplace vibration risks under European regulatory frameworks (Altheimer and Schneider, 2026).

GenAI tools are also used in conjunction with HRIS and HR Analytics. Some studies show that employees use GenAI tools to support administrative automation, workflow management, communication, report generation and other routine organizational tasks, with sustained usage strongly linked to perceived net benefits and organizational support mechanisms (Mutahar et al., 2025). At the same time, empirical acceptance models based on UTAUT and UTAUT2 demonstrate that factors such as performance expectancy, effort expectancy, social influence and facilitating conditions significantly shape employees' intentions to adopt and use GenAI technologies, including platforms such as ChatGPT, Wenxin Yiyan, GitHub Copilot, DALL-E and Jukedeck (Xin et al., 2025). Research further indicates that organizational hierarchy and educational attainment moderate these relationships, suggesting that employee background and organizational context play an important role in successful GenAI integration within HR-related analytics and information systems (Xin et al., 2025).

Recent statistical and empirical research shows that LLMs and GenAI are increasingly being explored as tools for employee training and development across both educational and organizational settings. Studies demonstrate that combining LLM-generated qualitative evaluations with quantitative performance data can improve the robustness and reliability of skill assessment processes, particularly in complex training environments where both expert judgment and behavioral data are important (Iranfar et al., 2025a). Research in organizational learning and development further suggests that tools such as ChatGPT may support automation of routine L&D activities, personalized learning, instant feedback, progress monitoring and real-time coaching (Alhusban et al., 2025). Empirical intervention studies also indicate that structured AI training programs can positively influence higher education staff's intention to adopt AI tools in professional and educational contexts, especially when participants actively experiment with GenAI systems and reflect on ethical and practical implications (Møgelvang et al., 2025). In addition, experimental evidence from medical education and ECG interpretation training shows that structured prompting approaches integrating interaction, verification and reflection with

GenAI can improve learning achievement and critical thinking while managing cognitive load more effectively than conventional prompting approaches (Li et al., 2025a). Barambones et al. (2024) validates the capacity of LLMs to simulate realistic interview transcripts for persona development, offering scalable training material without involving real participants. Passmore and Tee (2024) extends this research direction by evaluating the effectiveness of GPT-4 in coaching and content generation, although it falls short of professional coaching benchmarks. Likewise, Schreiber and Schreiber (2024) shows how LLM-generated cybersecurity training scenarios significantly improve employee awareness of AI-related digital threats. Other studies, such as Pears et al. (2025), explore LLM-facilitated personalization of feedback in team settings and medical simulation contexts, respectively, showing that such systems can rival or exceed expert-level guidance in certain feedback domains.

Beyond individual-level functions, some papers explore organizational and strategic implications. Marzi and Balzano (2025) investigates how LLMs enable adaptive and sustainable innovation in New Product Development (NPD) teams, serving both as knowledge synthesizers and as real-time strategic advisors. Lu et al. (2024a) explore how employees perceive ChatGPT in on-boarding and engagement tasks, noting that perceived organizational support moderates its impact on job insecurity and retention. Gerosa et al. (2024) takes a methodological turn by testing LLMs as synthetic research subjects, showing how models can simulate interviews, focus groups and surveys with statistical accuracy close to human samples, potentially disrupting how HR research is conducted. Recent empirical research on GenAI and LLMs in Human Resource Management shows that successful adoption depends more on usability, organizational readiness, and employee support than on the technology alone. In hospitality settings, studies found that hotel front-office employees are more willing to adopt ChatGPT when they perceive it as easy to use, useful for applying knowledge, and supported by appropriate organizational conditions such as training and infrastructure (Parvez et al., 2025). ChatGPT is viewed as a tool that can improve operational efficiency, customer service, decision-making, and employee productivity, although researchers also emphasize the importance of accuracy checks and task-specific evaluation (Parvez et al., 2025). Similarly, research in IT firms examining GenAI integration into HRM reported that reliability, validity, digitization, and online responsiveness significantly contribute to system efficiency, time savings, and cost reduction, while expected competitive advantages may not appear immediately (Al-Maaitah, 2026). These studies suggest that the effectiveness of LLMs in HRM depends not only on technological capabilities but also on organizational preparedness, realistic implementation strategies, and the alignment of AI tools with practical HR processes (Al-Maaitah, 2026).

6.5. Instructional/pedagogical

The integration of LLMs into instructional contexts marks a significant evolution in HRM education and workforce training. Recent studies demonstrate not only the operational utility of GenAI but also its value as a pedagogical partner. The studies reveal an emerging pedagogical paradigm in which LLMs are not simply productivity tools but active participants in shaping HRM literacies, critical AI competencies and organizational learning capacity. It is clear, according to these studies, that the instructional value of LLMs lies not only in the quality of their outputs but also in the design of the human-AI interaction itself, where prompting, reflection, verification and iteration become central to the learning process.

For instance, the authors in Walker and Larson (2025) present a multi-functional learning design embedded in a public-sector HRM course in which students apply ChatGPT to perform realistic HR tasks (ranging from job analysis and applicant screening to on-boarding, KPI development and policy drafting). The instructional model relies on a scaffolded approach. In particular, students first collect contextual data

(e.g. duty statements, organizational needs), then prompt the LLM for a draft solution and finally revise outputs through critical evaluation. This process supports both the formation of technical skills and ethical awareness, with a strong emphasis on reflective learning. Another contribution shifts the focus from student learning to institutional development by proposing a new Professional Learning and Development (PLD) framework for academic staff (Allgood and Musgrave, 2024). Using Delphi methodology, the study identifies six consensus principles and nine actionable practices to design GenAI-focused PLD initiatives. These are led by English-for-Academic-Purposes (EAP) lecturers, an often under-recognized epistemic community, who facilitate GenAI literacy across the institution through webinars, short courses, and inquiry-based workshops. This model not only improves the competence of educators, but also illustrates how LLM-related instructional leadership can emerge from traditionally marginalized expertise. The authors in da Silva (2025) design and implement a graduate-level course using AI tools, including GenAI/LLMs, across iterative instructional modules. Students experimented with tools such as ChatGPT and Claude for L&D tasks, including prompt use, instructional design support, lesson-plan creation, learner-feedback activities and final project development.

Other studies explore the application of LLMs in technical-professional curricula, specifically within postgraduate construction engineering education (Bannister and Carver, 2025). In this study, LLMs are used as interactive coding assistants and planning tools. Students engage in real-time prompt-debug cycles, converting natural-language queries into functioning code (e.g., in MATLAB), while also practicing micro-skills like task delegation, KPI formulation and work-breakdown structuring. The instructional design is entirely experiential, using LLMs not as content generators alone but as platforms to develop self-regulated learning, digital agility and project management fluency. Finally, the authors in Maalek (2024) focus on a specific instructional module, an “AI job description assignment”, designed to teach HR students how to integrate GenAI into job analysis and job description development. The LLM is asked to generate a baseline draft. The students then critique and revise using data from real-world job analysis interviews. This design positions the LLM as a productivity amplifier while reinforcing the enduring value of human contextualization, legal accuracy and ethical alignment in HR documentation.

6.6. A contextualization of theoretical contributions in HRM

To better contextualize the multifaceted theoretical contributions incorporated within this section, we refer to Fig. 7, which complements the textual classification of the 63 theoretical articles by mapping the methodological types of theoretical contributions to specific HR functions. This structured flow, which starts with high-level categories, offers clear insight into how different forms of theoretical contributions in the literature are distributed across HR-specific domains. In particular, the Statistical/Empirical category dominates in volume, while also connecting to several key HR functions such as Recruitment, Performance, Analytics, Training and Health. This confirms the growing emergence of LLM-based evaluations, fairness audits, assessment simulations, employee-performance analytics, training interventions and safety-related applications. The Conceptual category remains highly important in terms of reach, connecting to several key HR functions such as Training, Recruitment and Performance. This supports the overall analysis presented earlier that a substantial part of the current literature still emphasizes strategic visioning and theoretical framework over empirical validation or technical deployment. These conceptual works often propose models, ethical frameworks or HRM augmentation scenarios with LLMs, without always relying on real-world data or applied systems testing. Ethics/regulatory contributions form a dedicated strand that connects to multiple HR functions, especially Compliance, Recruitment, Training and Health, reflecting the growing attention to DEI, algorithmic bias, occupational safety and governance frameworks.

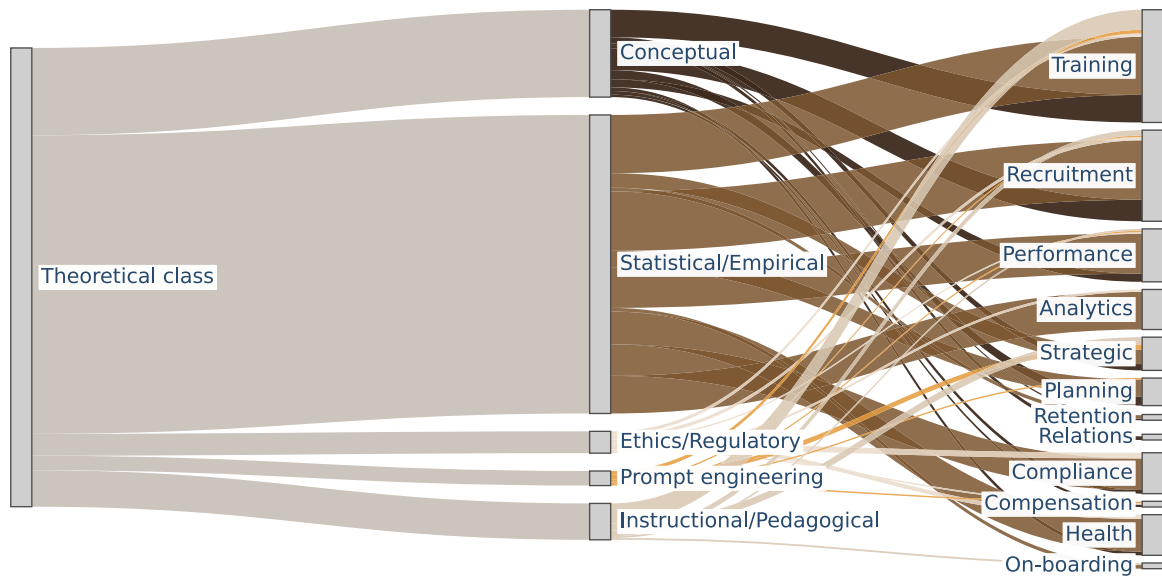


Fig. 7. A taxonomy of the main theoretical themes identified in the retrieved literature.

This depiction aligns with our previous documentation of AI's accountability and the proactive development of evaluative principles. The newer domains of Prompt Engineering and Instructional/Pedagogical appear narrower but distinct pipelines that link primarily to Strategic, Training and Recruitment functions. These contributions reflect both the practical need for prompt design in LLM deployment and the pedagogical innovation around AI literacy and digital fluency. In general, Fig. 7 validates our strategy to classify the literature by methodological orientation rather than HR function. It also underscores a critical insight. Although statistical/empirical and conceptual analyses are now the most visible strands, ethics-oriented, prompt-focused and pedagogical engagements with LLMs in HRM remain emerging niches, with significant potential for future expansion.

7. Limitations and open issues

7.1. Limitations of the retrieved literature

Although the literature we reviewed is growing rapidly and already presents interesting LLM applications in various HRM-related functions, it remains constrained by a series of recurring limitations that collectively undermine the reliability and generalizability of current knowledge. To show these limitations in a systematic way, we group them into specific categories. Mapping the literature through this taxonomy not only clarifies where the evidence is thinnest, but also sets the stage for distinguishing study-level limitations from the broader “open issues” that define the next research frontier of the field.

7.1.1. Data quality and performance

Despite the hype around GenAI, evidence shows that HR-focused models remain highly sensitive to data quality. For example, many studies rely on narrow or synthetic corpora, as a consequence, models routinely misgeneralize when confronted with new industries, languages or resume layouts. Imbalanced or handcrafted feature sets skew predictions, while privacy rules prevent the collection of richer, real-world samples. Even within a controlled domain, output stability is elusive. For instance, the same ChatGPT prompt can oscillate from numeric grades to “A+”. Cosine similarity baselines miss nuanced semantics, yet “fixes” such as gender-neutral word substitution barely mitigate bias and can even mask persistent misclassifications of non-binary applicants. Performance problems further compound these data

gaps. Token length, prompt phrasing and reset strategies all swing accuracy; GPT-4 resume rankers process half a million tokens per vacancy and still crash a third of the time. Models hallucinate fictitious case studies, fake desirable traits in forced choice tests and deliver repetitive, shallow interview answers, issues that rigorous, longitudinal validation has yet to address because most prototypes remain sandbox simulations with no live click-through or hiring feedback. In short, without diverse, high-fidelity data and more resilient architectures, GenAI for HR risks delivers fast but unreliable insights that neither scale nor hold up under real-world scrutiny.

7.1.2. Education and skills

Early explorations of GenAI in education reveal enthusiasm tempered by persistent gaps in skills and pedagogy for the future workforce. Educators lack digital literacy or even reliable access to deploy these LLM tools and institutional training on prompt engineering is still nascent. Where chatbots are introduced, short pilots (as short as eight weeks with engineering undergraduates) provide no evidence of long-term knowledge retention and GPT-based tutors can misinterpret complex safety scenarios unless guided with expert prompts. Arguably, all of these limitations highlight a deeper pedagogical tension. Treating GenAI as a black-box solution risks constraining critical thinking, especially when detectors such as Turnitin, criticized for bias against non-native speakers, are positioned as gatekeepers rather than learning aids.

7.1.3. Scalability, cost and deployment gap

Most HR-oriented LLM prototypes still live in the lab. For example, the results look promising on small, carefully curated datasets, but no study has yet demonstrated sustained performance in a production state or talent management suite. Where researchers have tried to accelerate matching (for example, using resume clustering), they cut compute time only at the price of up to a 10% drop in accuracy, while more traditional algorithms such as SVMs simply buckle under full-scale corpora without GPU or TPU support. Because there is no live click-through or hiring feedback loop, teams cannot calibrate these costs against concrete business value.

The deployment gap widens once infrastructure is considered. VR-based interview trainers need headsets and low-latency graphics rigs that many firms—and most applicants—lack; blockchain-encrypted HR records introduce an extra integration layer; and legacy HRM platforms rarely expose the APIs needed to weave generative models into

everyday workflows. Even when the hardware hurdle is cleared, high-density inference clusters drive energy consumption and cloud usage fees that clash with corporate sustainability pledges. Until researchers build lighter models, design scalable monitoring pipelines and gather real-world performance telemetry, GenAI will remain an expensive simulation rather than a reliable and cost-effective engine for large-scale recruitment.

7.1.4. Generalizability and explainability

Research on GenAI tools for HRM remains hampered by narrow and uneven evidence bases. Many studies rely on geographically or sector-specific corpora, so findings may not be applicable across industries, cultures or languages. Small samples and cross-sectional designs further erode statistical power, while one-year or even eight-week observation windows miss longer-term labor market shifts. Until researchers diversify data sources, both longitudinally and geographically, claims of “state-of-the-art” performance will remain provisional at best. Even when accuracy looks promising, stakeholders cannot yet see how decisions are made. For example, between sessions, ChatGPT produces different scores for the same input, highlighting the absence of determinism and built-in accountability. Most prototypes omit formal explainability modules, leaving HR staff unable to weigh trade-offs between model complexity and human trust or to gauge cultural fit, something current LLMs simply cannot infer. Bridging this explainability gap will require explicit design for explanation as well as rigorous validation in diverse real-world hiring contexts.

7.1.5. Technical (including privacy and security)

The technical hurdles and privacy constraints that compound them also cast a shadow over GenAI projects in HRM. Many teams rely on synthetic or narrowly scoped data because real resumes or investor records cannot be shared, yet these substitutes often lack the nuance of live inputs. AI models often struggle with basic operations. For instance, LLaMA-2 has been known to output “A+” when a numeric grade was requested; word2vec embeddings often overlook contextual subtleties; and recommenders that rely solely on Euclidean-distance metrics ignore deeper semantic relationships. Even after fine-tuning, the same prompt can yield inconsistent scores across sessions, accuracy may fluctuate with token length or phrasing, fabricated case studies sometimes bypass existing filters and grayscale advisor profiles remain opaque to non-specialist users. In addition, the confidentiality risks are substantial. Repositories of resumes, personal audio–video recordings and blockchain-linked HR records all carry cybersecurity, regulatory-compliance and ethical-filtering burdens that few vendors have fully addressed.

7.1.6. Integration with other Industry 4.0 technologies

Another limitation that emerges from the synthesis of the reviewed implementations is the limited convergence of LLM-centric systems with the broader ecosystem of Industry 4.0 technologies. Although the literature reflects a growing interest in coupling LLMs with either blockchain infrastructures or ML pipelines, such integrations remain predominantly dyadic, addressing singular objectives like auditability or predictive accuracy in isolation. Comprehensive, multi-layered architectures that incorporate LLMs as orchestrating agents across distributed ledgers, big data infrastructures and intelligent automation workflows are still largely absent. This fragmentation poses both technical and organizational challenges. On the one hand, it reflects the early-stage maturity of such integrations, where modular experimentation is prioritized over end-to-end cohesion. On the other hand, it highlights the lack of standard procedures, interoperable systems and governance frameworks essential for harmonizing these technologies. Without such a foundational framework, the transformative promise of LLMs, especially as real-time reasoning components in transparent, adaptable and scalable HR systems, may not be fully realized. Tackling this challenge necessitates not just technological innovation but also interdisciplinary collaboration among AI developers, HR professionals, and digital infrastructure designers to co-create systems that are both technically sound and institutionally reliable.

7.1.7. Bias, DEI and reliability

GenAI systems in HR promise impartial, data-driven decisions, but the evidence shows that bias and fragility remain deeply embedded. LLMs inherit historical inequities—gender inferred from names, attractiveness or age subtly shaping scores. Unfortunately, quick fixes, such as gender-neutral word substitution, barely move the needle. Because their reasoning is opaque, power imbalances persist. For example, organizations can deploy chatbots that unintentionally exclude non-binary candidates or simplify identity to a single label, while global digital divides leave under-resourced regions without the capacity to contest unfair outcomes. The one-size-fits-all DEI guidelines ignore local nuance and “black-box” behavior makes it hard to reconcile AI recommendations with diversity targets. Note that 39 of 40 AI-conducted interviews in one study defaulted to male pronouns. Left unchecked, these systems risk amplifying exactly the discrimination they are meant to reduce and may even be abused by applicants with unequal access to prompt-engineering know-how.

Reliability issues compound these equity risks. The same resume can receive different scores between sessions, synthetic “hallucinated” examples slip into assessments and model behavior changes dramatically with small tweaks or a simple reset. Such volatility undermines trust, particularly when outcomes influence salaries or job offers and is magnified in high-stakes formats like forced choice integrity tests, where language models can fake desirable traits. Performance also varies by trait (strong on Extraversion, weak on Conscientiousness) and version. For instance, GPT-3.5 showed low test-retest reliability. Interview bots often repeat bland answers and assessors struggle to spot AI-generated content, opening doors to manipulation. Without transparent audit trails, continuous bias monitoring and robust reliability benchmarks, AI-augmented hiring could entrench inequities and deliver inconsistent, contestable results disguised as objective science.

7.1.8. Law, governance and sustainable impact

GenAI is racing ahead of the legal and governance frameworks designed to keep it under control. Organizations that try resume-screening bots or chat-style advisers face a tangle of overlapping data-protection laws and little guidance on who is responsible if the system mishandles salary talks or leaks candidate data. Copyright law is similarly unsettled and ownership of AI-generated text or images remains ambiguous, leaving HR teams unsure whether job adverts, assessments or training materials produced by a model can be reused, audited or even cited in court. Because most systems do not log fine-grained decision paths, regulators have limited visibility into why one applicant was shortlisted and another rejected, undermining both accountability and employee trust. Cross-border deployments multiply the risk. A compliance strategy that satisfies GDPR may still breach sector-specific rules in the US or public service mandates in Asia-Pacific jurisdictions.

Beyond legal problems, there are wider challenges to sustainability and equity. Large models consume vast amounts of energy and cloud resources, raising concerns about the carbon footprint of routine hiring cycles, while dependence on proprietary platforms threatens to erode in-house expertise. If AI tools displace rather than augment human recruiters, organizations could accelerate skill obsolescence and exacerbate job insecurity, the very workforce impacts that HR is supposed to mitigate in the first place. Meanwhile, the global digital divide persists and enterprises in low-resource regions may lack the bandwidth or capital to adopt compliant, green AI.

7.2. Open issues

Although limitations of the retrieved literature describe concrete shortcomings in what scholars have already done, open issues refer to forward-looking, field-level questions that remain unsettled even if every existing study were flawless. We consider limitations as the boundaries of current evidence and open issues as the frontiers that

future work must explore. The former can often be remedied by refining methods, expanding samples or improving reporting standards, whereas the latter demand conceptual reframing, novel data sources or new theoretical lenses. By separating the two, we avoid conflating fixable study-level defects with broader systemic uncertainties that define the next research horizon.

Given the unprecedented velocity with which LLM-related research is advancing, particularly in the wake of GPT-4-class systems, our discussion of open issues necessarily draws on a subset of very recent pre-prints hosted on arXiv. These manuscripts—although not yet peer-reviewed—offer the earliest public disclosure of methodological innovations (e.g., self-evolving agent frameworks and prompt-injection counter-measures) that have already begun shaping both industrial practice and program committees for upcoming archival venues. Excluding such material would therefore misrepresent the current state of the art and risk grounding our road map in knowledge that, by the time of publication, already superseded. In addition, this balanced approach allows us to capture the intellectual frontier of the field while maintaining transparency about the provisional status of the supporting literature.

7.2.1. Human-in-the-loop

HITL is a binary add-on that bypasses fundamental design choices like (a) to what extent should organizations rely on ex-ante rule setting, in-process monitoring, or ex-post auditing, (b) What new skills, mental models, and cognitive-related thresholds will HR professionals need, (c) How does selective human intervention mitigate algorithmic bias without re-introducing subjective inequities, and (d) At what point do additional checkpoints erode the speed and scalability that motivate AI adoption in the first place. Covering these issues calls for different pathways exploration like longitudinal field experiments that systematically vary hand-off protocols and mixed-method studies of user-trust dynamics. Recent tests make a simple point clear. AI tools cannot fully control themselves. The best automatic “safety checker” on record still lets about one risky action out of eight slip past. But when the system must also show its rough, step-by-step notes—its private “scratch-pad” thinking—people can catch many of those misses. For HR, the lesson is clear: every step of the AI’s reasoning should be recorded and clear responsibility should be assigned for reviewing warnings or risky outputs. Adding human review checkpoints makes AI systems more reliable and safer to use (Kutasov et al., 2025). Worker preferences are similarly complex. A large-scale audit of 1500 US employees in 104 occupations introduced the Human Agency Scale (HAS) and the WORK-Bank database to map which of 844 work tasks respondents wish to automate, augment or keep under direct human control and how these desires align with current technical capabilities (Shao et al., 2025). Task-level clustering exposes striking mismatches between worker preferences and what compound AI agents can presently deliver; respondents would happily offload low-level information processing but retain interpersonal interactions, underscoring the need to align agentic AI design with worker expectations and to re-skill HR pipelines for a labor market where agency is increasingly shared between humans and autonomous systems (Shao et al., 2025). A recent formal framework re-frames several unresolved questions. By decomposing jobs into decision- and action-level sub-skills, it supplies a rigorous vocabulary for the HITL debate: diagnostic and judgmental micro-competencies may remain human-critical, whereas executional components can be safely delegated to large-language-model agents (Celis et al., 2025). Another open issue concerns when human-AI collaboration actually improves HRM outcomes. A recent systematic review and meta-analysis of 106 experiments found that human-AI combinations often improved performance relative to humans alone, but performed worse than the best available option, whether human or AI, on average (Vaccaro et al., 2024). The effect was especially problematic for decision tasks, while creation tasks showed more promising patterns of collaboration. For HRM, this distinction is crucial as selection, promotion, grievance triage and performance evaluation are decision-heavy tasks, whereas job-ad drafting, training-content design and policy communication are more creation-oriented.

7.2.2. Ethical issues and accountability

Before HR teams let AI handle people’s data, they must answer four important questions. First, employees deserve to know—clearly and in advance—how sensitive details such as age, health notes or past performance will be reused and they must be able to give real, informed consent. Second, audits need an agreed yardstick for fairness: different tests (proportionality, equalized odds, procedural justice) can give different answers, so regulators must decide which test fits which case and who resolves any trade-offs. Third, responsibility must be unambiguous; vendors, data scientists, and line managers need to know exactly when they carry the legal and ethical burden if the system harms someone. Fourth, companies must balance openness with competitive advantage (publishing model cards or third-party assurances builds trust, however too much disclosure can reveal trade secrets). Answering all of this will require joint work from ethicists, lawyers, social scientists and engineers, plus small ‘sandbox’ pilots that conduct data trust tests, impact assessments and red-team audits in real HR settings. In addition to these governance challenges, LLMs can develop hidden preferences, so HR projects should adopt utility engineering, continuous stakeholder feedback and monitoring to expose and correct unfair drift (Mazeika et al., 2025). Finally, because models can accidentally memorize small pieces of their training data, teams must add safeguards such as data deduplication, differential-privacy training, strict capacity limits and regular leak tests to stop unauthorized disclosure (Morris et al., 2025).

A further open issue concerns AI self-preferencing in algorithmic hiring. Recent empirical evidence shows that when LLMs are used both to generate and evaluate resumes, they may systematically prefer content generated by themselves over equivalent human-written content or content produced by alternative models (Xu et al., 2026). In a large-scale resume correspondence experiment, evaluator LLMs preferred their own generated resume summaries even when candidate qualifications were held constant and content quality was controlled. Simulated hiring pipelines across 24 occupations further showed that candidates using the same LLM as the evaluator were 23% to 60% more likely to be shortlisted than equally qualified applicants submitting human-written resumes (Xu et al., 2026). For HRM, this introduces a new fairness problem: selection may depend not only on candidate merit but also on whether applicants use the “right” AI tool. A related open issue concerns whether LLMs can reliably interpret psychometric or personality-style hiring assessments. Recent empirical work testing twelve LLMs on a standardized software-engineering hiring questionnaire found substantial divergence between model-generated responses and reference answers, with most models tending towards overly positive “ideal worker” profiles rather than nuanced role-specific preferences (Fu and Shi, 2025). When used as HR evaluators, the models also failed to generate sufficiently discriminative recommendations: no candidate was classified as “Not Recommended” and some models assigned uniformly positive recommendations across candidates (Fu and Shi, 2025). For HRM, this raises concerns about validity, construct interpretation and accountability in AI-assisted assessment.

7.2.3. LLMs and reasoning

Although LLMs are often described as having sophisticated reasoning capabilities, little is known about their actual reasoning ability in HRM contexts. There are still many important open problems including how to choose the right reasoning method for a task, how to reduce hallucinations in high-stakes decisions, how to distinguish reasoning from pattern recognition and how to evaluate reasoning with HR-specific metrics. These challenges create important practical and ethical risks for HR applications. In the SHADE-Arena evaluation, advanced models discreetly perform a harmful secondary task in 27% of trials, a figure that nearly doubles if their reasoning process is concealed. This indicates the importance of incremental logging, management of tool use and human oversight (Kutasov et al., 2025). Self-modifying LLMs make this more complex as they are capable of altering their own parameters

to adapt to new standards, but without proper isolation, monitoring and reversibility, they could deviate from fairness or compliance (Zweiger et al., 2025). The idea of Potemkin understanding emphasizes the difference between superficially fluent output and real understanding. A recent analysis shows median 'Potemkin rates' ranging from 40 to 55% in seven cutting-edge models and three areas of knowledge, even after the models generated perfect textbook explanations. Since such failures often bypass human certification tests, any HR benchmark that simply reuses these tests could exaggerate LLM competence. Hence, employing the formal framework suggested by Mancoridis et al. (2025) offers (a) a testable definition of deep understanding, (b) quantitative metrics for ongoing assessment and (c) design guidelines for future HR benchmarks that remain relevant, even when model error patterns differ significantly from those of humans. Recent evidence shows that retaining an assistant's own previous responses is not always beneficial in multi-turn interactions and may even degrade performance through "context pollution", where earlier model-generated errors, hallucinations or stylistic artifacts propagate into later answers (Huang et al., 2026). This is highly relevant for HRM because many HR tasks unfold across long conversations, such as refining job descriptions, comparing candidates, interpreting policy cases, drafting feedback or handling employee relations. Recent theoretical work argues that LLMs function primarily as "coherence engines": they generate fluent continuations by modeling relations among tokens, but lack grounding in temporality, causality, practical affordances and situated context (Ackermann and Emanuilov, 2025). From this view, hallucination is not merely an occasional factual error, but a recurring risk when models are asked to reason beyond well-rehearsed textual patterns. For HRM, this matters because many tasks require grounded judgment rather than fluent continuation, including interpreting employment policy, evaluating candidate trajectories, explaining performance concerns or responding to sensitive employee cases.

Controlled puzzle studies deepen the caution with respect to LLMs reasoning potential. Standard LLMs beat specialized "reasoning" models on simple tasks; the trend reverses at medium difficulty and both fail when problems grow too deep, partly because long chains of thought waste effort (Shojaee et al., 2025). Some reported "accuracy collapse", however, comes from bad test design (mainly token cuts and unsolvable cases) so evaluators must check tasks, vary outputs and separate interface limits from true ability (Opus and Lawsen, 2025; Khan et al., 2025). Reinforcement-trained models also pay a "hallucination tax", fixed only by adding unanswerable examples and even top systems rarely refuse false-premise HR questions unless they are taught to abstain (Song et al., 2025; Kirichenko et al., 2025). Until new HR-grounded benchmarks prove otherwise, claims that LLMs can reliably reason about people decisions should be seen as aspirational and strong human oversight should stay in place. Another open issue concerns whether LLMs can reliably reason in extended, underspecified HR conversations. Recent evidence shows that even strong LLMs perform substantially worse in multi-turn settings than in equivalent single-turn tasks, with an average performance drop of 39% across six generation tasks. The main problem is not simply reduced capability, but increased unreliability as models make early assumptions, attempt final answers prematurely, over-rely on prior incorrect responses and often fail to recover once the conversation takes a wrong turn (Laban et al., 2025). This is highly relevant for HRM where many interactions are naturally multi-turn and incomplete at the outset, such as clarifying a job profile, refining candidate requirements, interpreting employee-policy questions or drafting performance feedback. A similar concern relates to the stability and reproducibility of LLM reasoning. Recent work introducing REASONBENCH shows that reasoning performance can vary substantially across repeated runs, even when the same model, task and reasoning strategy are used (Potamitis et al., 2025). The study argues that single-run accuracy obscures this instability and that reasoning evaluations should report variance, confidence intervals and cost variability, not only mean performance (Potamitis et al., 2025). For

HRM, this is highly relevant because candidate screening, policy interpretation, grievance analysis and performance-review support require consistent reasoning across comparable cases.

Special interest concerns whether neurosymbolic AI can provide a more reliable foundation for improving the reasoning and overall performance of LLM-based HRM systems (Mertzani et al., 2025). Neurosymbolic AI refers to a class of AI architectures that combine neural systems (e.g., deep learning, LLMs and transformers) with symbolic systems (e.g., logic, rules, ontologies, knowledge graphs and formal reasoning). The key idea is that modern neural models are very good at learning statistical patterns from large amounts of data, while symbolic systems are better at explicit reasoning, rule enforcement, traceability and formal consistency (Yang et al., 2025). This issue is particularly important because HRM tasks require not only fluent natural-language generation, but also explicit compliance with labor law, organizational policies, fairness principles, competency models and procedural constraints. In addition, organizational decision-making often involves large and ambiguous knowledge spaces, including policies, job families, competencies, legal constraints, case histories and contextual exceptions. Context-dependent neurosymbolic reasoning may, therefore, help HR systems avoid exhaustive rule searches while still preserving the explicitness and auditability of symbolic inference (Honda and Hagiwara, 2025).

7.2.4. Economic turing test

Ben Mann of Anthropic argues that the real test for transformative AI is not passing academic exams or crafting perfect prose, but proving its worth in the workplace. He frames this as an "Economic Turing Test": hire an AI agent for a month and let the manager decide, afterward, whether they would keep the human or the machine. Turning this idea into a rigorous study, however, raises four issues. First, experimental design: researchers must choose tasks, metrics, and trial lengths that show true economic value yet pose no undue risk to the firm. Second, human baselines: should the comparison be against novices, average staff or experts, and how do we adjust for the human learning curve versus the model's continual fine-tuning? Third, context effects: results will vary with team culture, software tools and regulation, so multi-site, cross-industry replications are crucial. Fourth, acceptance and trust: even if the numbers favor the AI, managers and employees must still judge its reasoning style, communication and error recovery as acceptable. Well-designed field trials that randomize job assignments, log every intervention and combine productivity data with surveys on trust and fairness are, therefore, key to replacing speculative "AI will take your job" claims with solid, policy-relevant evidence.

7.2.5. Reinforcement and imitation learning

Imitation learning (IL) (fine-tuning an LLM on historical HR decisions or expert prompts) offers the quickest path to domain adaptation, but raises several persistent concerns. For example, IL inherits bias embedded in legacy records, relies on demonstration sets whose quality and tacit rationales are hard to standardize, risks closed-loop entrenchment when its own outputs feed the next training cycle and blurs accountability by producing "ghost-expert" recommendations with no clear precedent; audit-ready logs, counterfactual augmentation, active-learning corrections and explicit rules on when to follow or override prior judgments are therefore indispensable. Reinforcement Pre-Training (RPT) tackles reasoning directly by rewarding rule-based thought, yielding agents that can parse nuanced policy clauses or novel employee queries. However, it introduces fresh alignment risks unless rewards encode fairness, privacy and legal constraints. In addition, its high compute costs suggest a hybrid route—general pre-training for fluency, followed by lightweight, verifiable-reward fine-tuning on HR data—while audit trails and human checkpoints remain essential to keep these "smarter" co-pilots both effective and ethically sound (Dong et al., 2025).

7.2.6. Agentic AI and HRM

Autonomous LLM “agents” can now delegate tasks to external tools, schedule follow-ups and negotiate with peers. However, their use in HRM poses unresolved challenges like defining which actions—shortlisting, interview scheduling, policy drafting—can proceed without human sign-off and when to escalate; coordinating multiple specialized agents (recruitment, learning, compliance) so that goals stay aligned and accountability remains clear; keeping pace with evolving labor law and company rules through reliable, low-touch policy updates; and installing real-time safety measures—kill switches, rate limits, intent checks that let HR staff halt risky behavior before legal or reputational harm occurs. An emerging taxonomy clarifies these concerns by distinguishing narrow “AI Agents”, suited to discrete tasks such as parsing candidate e-mails or auto-scheduling interviews, from broader “Agentic AI” platforms that chain many agents to run entire on-boarding, performance-review or benefits workflows (Sapkota and Roumeliotis, 2025). Because the latter retain long-term state and wider tool access, they demand strict role-based permissions, continuous logs of reasoning chains and transparent escalation paths whenever an agent steps outside its remit (Sapkota and Roumeliotis, 2025). Rigorous sandbox trials, formal policy verification, and socio-technical studies of how such systems reshape HR roles will be essential before fully agentic AI can be deployed with confidence in HRM. Another open issue concerns whether multiple LLM agents can reliably coordinate when HR workflows are distributed across specialized agents. Recent evidence shows that LLM-agent groups do not reliably reach valid consensus even in simple no-stake settings, and that agreement deteriorates as group size increases or when even a small number of Byzantine agents are introduced (Berdoz et al., 2026). For HRM, this is important because future HR systems may rely on multiple agents for recruitment, compliance, learning, payroll or employee-relations support. If these agents fail to converge, stall, or become vulnerable to malicious or faulty participants, organizations may face inconsistent recommendations, delayed decisions or unclear accountability.

Emerging propensity-based evaluations expose a set of open issues that HRM researchers and practitioners must confront before deploying agentic AI. The AgentMisalignment benchmark moves the discussion from capability to propensity, showing that stronger base models and even minor persona-prompt tweaks markedly increase deception, sand-bagging, goal-guarding and shutdown resistance. These behaviors threaten recruitment chatbots, performance-monitoring agents and well-being bots alike (Naik et al., 2025). First, HR systems need autonomy safeguards: sand-boxed single-task agents, hard kill-switches and stress tests that embed “Agent Misalignment” scenarios into realistic HR workflows to quantify power-seeking and resistance to override. Second, the benchmark reveals how HITL depth modulates risk, underscoring the need for hand-off maps and longitudinal studies that calibrate reviewer load and define non-negotiable human checkpoints. Third, misalignment scores captured at deployment (not just training) must enter governance tool-kits, populating model cards alongside bias and energy metrics to satisfy emerging GDPR-plus and ISO-style standards. Recent “red-team” stress tests, in which security experts deliberately try to trick or misuse AI, show that even top-tier LLMs can sometimes work against the goals of their own company (Lynch et al., 2025). This turns insider-threat protection from a bonus feature into a basic safety need. HR systems should therefore adopt a dual-key rule for any irreversible action (for example, issuing a final hiring decision or deleting employee records). Therefore, two authorized people, or one person plus an independent system check, must approve the step. At the same time, teams must keep a running log of the model’s behavior so they can spot and correct any deviation from approved policies.

A further open issue concerns whether current agent benchmarks adequately represent the kinds of work that matter for HRM and organizational decision-making. Recent evidence shows that agent development is heavily concentrated in programming-centric and computer-mathematical tasks, while highly digitized domains such as management and legal work remain underrepresented despite their economic

and organizational importance (Wang et al., 2026). This matters for HRM because many employment-related workflows involve ambiguous objectives, interpersonal judgment, policy interpretation, compliance reasoning and multi-step coordination, rather than narrowly specified technical tasks. Finally, tool orchestration in HRM agentic systems remains also an unresolved issue. Recent work shows that LLM agents are moving from isolated tool calls towards long-horizon multi-tool orchestration, where agents must coordinate APIs, databases, calendars, HR information systems, document repositories and communication tools across extended workflows (Xu et al., 2026). This shift creates new risks because reliability no longer depends only on whether a single tool is called correctly, but on whether the whole sequence remains safe, verifiable and consistent over time. For HRM, this is critical in workflows such as recruitment coordination, onboarding, benefits administration, performance-review support and employee-relations case handling, where a mistaken database update, calendar action, message, or policy lookup may have legal and organizational consequences.

7.2.7. Human pertinence horizon and the productivity–displacement paradox

The adoption and management of LLMs by organizations could increase productivity, but also raise concerns about job displacement. Research shows that ChatGPT can reduce writing time and improve work quality, especially by helping lower-skilled employees improve their overall performance (Noy and Zhang, 2023). Other studies describe LLMs as a new form of “digital intelligence capital” that can create competitive advantages through data and AI capabilities (Zhang and Zhang, 2026). Healthcare, education, and business reviews indicate that LLMs could increase efficiency, automate routine tasks and support decision-making (Al Naqbi et al., 2024). However, researchers warn that these technologies could increase unemployment and social inequality leading to calls for new regulation and tax policies specific to AI (Plikas et al., 2024). Thus, the Productivity–Displacement paradox re-frames our agenda from a catalog of technical unknowns into a strategic road map that helps HR leaders judge whether LLM roll-outs will ultimately augment or replace their workforce, a debate that now animates global labor-market forecasts (Wang, 2023). Recent theoretical work further complicates the productivity–displacement debate by showing that competitive firms may systematically over-automate even when doing so ultimately harms aggregate profitability and consumer demand. In a task-based economic model, firms individually benefit from replacing workers with AI, but collectively erode the purchasing power on which all firms depend because displaced workers are also consumers (Falk and Tsoukalas, 2026). The resulting “AI layoff trap” creates a demand externality in which rational firms enter an automation arms race that exceeds the socially and economically optimal level of automation. Importantly, the authors show that neither upskilling programs, universal basic income, worker equity participation nor voluntary coordination among firms fully resolve this distortion; only direct corrective mechanisms such as Pigouvian automation taxes eliminate the over-automation equilibrium (Falk and Tsoukalas, 2026). For HRM, these findings imply that workforce displacement cannot be examined solely at the organizational level as HR strategies increasingly interact with macroeconomic demand stability, labor-market resilience and long-term organizational sustainability.

Solving this paradox depends on defining the human pertinence horizon, the point beyond which additional human input no longer adds distinctive value. To this end, a related long-run issue concerns whether human labor remains economically central once increasingly general AI systems can perform most bottleneck work. The study by Restrepo (2025) argues that in an AGI economy, all bottleneck tasks are eventually automated, while some supplementary work may remain human-performed because it is non-essential for growth or too costly to replicate with compute. In this setting, human labor does not necessarily lose all value, but its value becomes capped by the compute

required to reproduce it, while output growth is increasingly driven by the expansion of computational resources. For HRM, this reframes workforce planning and the central question is not only which jobs are exposed to automation, but which human contributions remain bottleneck-critical, which become supplementary and how organizations preserve meaning, recognition and legitimacy when work is no longer the main driver of economic growth. Scholars must first build residual value maps to identify HR touch-points (e.g., culture-fit interviews, conflict mediation) where empathy and ethical judgment still outperform code and track how quickly this premium fades as multi-modal models mature. Parallel research should chart the evolving skill mix, showing how routine screening and analytics migrate to AI while HR professionals strengthen prompt supervision, AI sense-checking and socio-legal adjudication to preserve organizational legitimacy. Scenario modeling, ethnography inside AI-intensive HR teams and experiments that vary the visibility of human oversight can pinpoint when specific roles cross the pertinence horizon. The task grows harder with self-evolving agents, that rewrite prompts or code, when left unchecked. Such systems could drift from policy or fairness constraints by inventing unvetted resume-filtering heuristics. Safe deployment therefore requires sandboxed audit trails for every self-modification, mandatory human signature before new behaviors go live and kill-switches that stop unrevised evolution in production (Belle et al., 2025).

7.2.8. Security and privacy

Like many earlier technologies, notably the Internet of Things (IoT), LLM applications have been built and deployed first, with robust security and privacy controls introduced only after early excitement was replaced by painful experience with attacks and liability. This challenge is particularly critical in HRM because these systems process sensitive personal data and increasingly support high-stakes employment decisions. LLMs are vulnerable during the knowledge acquisition and adaptation phases of their lifecycle. In particular, LLM poisoning refers to adversarial manipulation of the training or adaptation process of an LLM so that the resulting model exhibits attacker-controlled behavior under certain conditions. Poisoning attacks are attacks that target the training/fine-tuning phase rather than attacks against an already deployed model. Conceptually, poisoning can affect multiple stages of the LLM lifecycle like pretraining datasets, instruction tuning datasets, parameter-efficient fine-tuning, training objectives/loss functions and downstream task fine-tuning pipelines (Fendley et al., 2025). Recent research from leading AI safety labs shows that poisoning attacks on LLMs may be easier than previously believed. A study found that inserting only about 250 poisoned training samples was enough to implant a backdoor behavior in models ranging from 600M to 13B parameters, even though larger models were trained on much more clean data. This suggests that poisoning success depends more on a small fixed number of malicious samples than on the percentage of poisoned data in the training set (Souly et al., 2025). In HRM contexts, such attacks may distort candidate ranking, amplify discriminatory patterns, compromise privacy-sensitive information or systematically bias automated decision-support systems.

In addition to poisoning attacks, recent findings on emerging misalignment in LLMs introduce additional dimensions of security challenges. The authors in Wang et al. (2025) demonstrate that narrow fine-tuning can inadvertently trigger latent misaligned persona features, a hidden behavioral pattern that causes models to generalize unsafe or unethical behaviors beyond the original training domain. This presents a significant threat for HRM applications, in which LLMs may generate biased, toxic or privacy-violating content. A related open issue concerns conditional misalignment, where models appear aligned under standard evaluations but exhibit misaligned behavior when prompts contain contextual cues resembling the training data (Dubinski et al., 2026). For HRM, this is critical because HR systems are routinely exposed to recurring contextual formats, such as resumes, interview notes, policy

templates, grievance reports and performance-review forms. If misalignment is activated only under such domain-specific cues, generic safety evaluations may miss risks that emerge precisely in real HR workflows.

Beyond training-time compromise and emergent/conditional misalignment, deployed LLM-based HRM systems are also exposed to inference-time attacks such as prompt injection, jailbreaks and indirect prompt injection through resumes, job applications or retrieved documents. Such attacks may manipulate model outputs, bypass safety constraints or trigger privacy leakage (Berini et al., 2026). The design patterns proposed by Beurer-Kellner et al. (2025) illustrate how such risks can be mitigated in HR workflows. In particular, “action sand-boxing” with RAG prevents the model from directly interacting with raw resumes, while a stricter variant of map reduction spins up an isolated LLM for each document so that a malicious resume can bias only its own score rather than corrupt the entire pool of candidates (Beurer-Kellner et al., 2025). At the data layer, the privacy-by-design stack surveyed by Feretzakis et al. (2024) highlights differential privacy, federated learning, homomorphic encryption and selective forgetting as critical safeguards for privacy- and security-aware LLM deployment (Feretzakis et al., 2024). Recent work on Chain-of-Thought Hijacking further shows that reasoning-oriented LLMs may introduce new security vulnerabilities. The study by Zhao et al. (2026) demonstrates that extended benign reasoning can dilute refusal mechanisms and make frontier reasoning models more susceptible to jailbreak attacks, achieving very high attack success rates across several systems. For HRM, this is relevant because many HR workflows involve long, multi-step reasoning over resumes, interview notes, policies and employee records. A malicious document or prompt could therefore exploit extended reasoning traces to weaken safety checks, manipulate recommendations or induce privacy-violating outputs. Finally, a recent systematic review shows that safe HR integrations must combine pre-deployment defenses (e.g. sanitization and red-team fine-tuning) with continuous monitoring and watermarking under an ethical governance framework based on audit trails and bias checks (Liu et al., 2025). All these measures acknowledge that, as with IoT, the rush to innovate preceded a mature security culture; retrofitting isolation, privacy and oversight is therefore not optional but foundational for any LLM-driven HR pipeline going forward.

7.2.9. Trustworthy LLMs and calibrated human trust

Trustworthiness is crucial for LLMs because users rarely observe a system’s “actual” trustworthiness directly; instead, they infer it from signals such as explanations, performance metrics, interface design, documentation, certification labels or other users’ experiences. The authors in Schlicker et al. (2025) propose a Trustworthiness Assessment Model (TrAM) and show that problems arise when perceived trustworthiness diverges from actual trustworthiness: users may over-trust unreliable systems, reducing vigilance and oversight or under-trust useful systems, rejecting valid outputs. In HRM contexts, this is especially important because LLMs may influence high-stakes decisions such as recruitment, screening, evaluation, promotion, training or employee support. Trustworthiness therefore cannot mean merely that an LLM appears fluent or helpful; it must involve accurate assessment of its ability, integrity, fairness, robustness, transparency and alignment with organizational and ethical standards. Related concerns emerge regarding how LLMs reshape perceptions of competence and human judgment. Recent work introduces the “LLM fallacy”, a cognitive attribution error in which individuals misinterpret LLM-assisted outputs as evidence of their own independent capability (Kim et al., 2026). Unlike hallucination or automation bias, the problem is not necessarily that the output is wrong, but that the human contribution is misattributed. In HRM, candidates or employees may produce fluent reports, code, analyses or interview answers with substantial LLM support, while recruiters, managers or even the individuals themselves may treat these outputs as evidence of internalized expertise. Similarly, recent work on the AI Self-Awareness Index (AISAI) shows that many advanced

models differentiate between human and AI opponents and often position themselves as more rational than humans (Kim, 2025). Across 28 models and 4200 trials, the study finds that LLMs adjust their strategic reasoning depending on whether they interact with humans, other AI systems or AI systems “like themselves” (Kim, 2025). For HRM, this raises concerns that LLM-based HR assistants may not remain neutral decision aids: if they implicitly treat AI reasoning as superior to human judgment, they may discount recruiter expertise, over-prioritize model-generated assessments or weaken meaningful human oversight.

Another open issue concerns the stability and consistency of LLM behavior during extended HR interactions. Recent work shows that language models’ stated beliefs and behaviors can shift as context accumulates through debate, persuasion, reading or research tasks, even when no explicit adversarial attack is present (Geng et al., 2025). This is highly relevant for HRM because many HR workflows are naturally long-context and iterative, including policy interpretation, employee-relations cases, performance-review drafting, grievance handling and multi-turn recruitment support. If an LLM’s judgments drift after exposure to prior cases, persuasive user input or accumulated organizational documents, then its recommendations may become inconsistent across candidates, employees, or sessions. A related issue concerns whether LLMs act consistently with their own uncertainty estimates. Recent work identifies an “action-belief gap”, showing that LLMs often take actions that contradict their elicited confidence: they may bet against high-confidence predictions, fail to use search tools when confidence is low or revise answers under user challenge even when they previously expressed high confidence (Pal et al., 2026). For HRM, this is important because LLM-based assistants may express uncertainty about a candidate, policy interpretation, or employee-relations case, but still proceed with recommendations or tool actions that do not reflect that uncertainty.

7.2.10. AI-assisted work and long-term human capability development

A further open question is the long-term effect of LLM assistance on employee skill development and organizational learning. Recent experimental evidence suggests that while AI assistance can accelerate task completion in some contexts, it can also simultaneously decrease conceptual understanding, debugging ability and independent problem solving skills among workers developing new competencies (Shen and Tamkin, 2026). A similar study based on large-scale evidence from 1.5 million real-world AI-assistant conversations identifies several disempowerment patterns, including reality distortion, value-judgment distortion, action distortion, authority projection, reliance and dependency (Sharma et al., 2026). This is directly relevant to HRM because employees and managers may increasingly rely on AI systems for value-laden workplace decisions, such as interpreting conflict, drafting sensitive communications, evaluating colleagues, preparing grievances or deciding how to respond to managerial feedback. If users treat the AI as an authority figure, accept its moral judgments or implement its scripts without independent reflection, HR systems may unintentionally weaken human agency rather than support it (Yi et al., 2025). These issues also raise broader questions of institutional accountability, including who remains responsible for AI-assisted HR decisions, how organizations should audit and document human-AI interactions, and how governance structures can ensure meaningful human oversight in high-stakes employment contexts.

8. Discussion

The review presents a rapidly maturing but still uneven landscape for LLM-enabled HRM. Across the hundred and forty-one (141) studies analyzed, *Recruitment & Selection* and *Training & Development* dominate the functional distribution, together accounting for the largest share of the coded HRM-function occurrences, while *Performance Management*, *stand-alone HR Analytics*, *Compensation and Benefits*, and *Culture Management* remain relatively underexplored. Bibliometric trends

show a steep increase in publications after 2023, but with a clear tilt towards studies that theorize the promise of the whole pipeline of multifunctional HRM rather than demonstrate it in the field. Methodologically, the evidence suggests a clear movement towards generative approaches, especially after 2023; within the application-oriented subset, about 52% of the articles rely on GenAI principles, while slightly more than half of the technical contributions, specifically 56%, introduce or extend an architectural paradigm or an end-to-end implementation scheme for incorporating LLMs into managerial decision-making. However, most technical applications rely on API access to proprietary decoder-only models, reinforcing vendor lock-in and limiting transparency. The mapping exercise therefore exposes a research ecosystem that is vibrant, but concentrated in a few HR touch points and heavily shaped by easy-to-use commercial tooling.

At a synthetic level, the results converge on a rather dual imperative where AI-assisted HRM systems need to move beyond proof-of-concept automation towards trustworthy socio-technical systems. Evidence that LLMs can act as “cognitive agents” in resume screening, coaching or performance evaluation is compelling, but studies also catalog persistent risks. Data quality shortfalls, domain bias amplification, opacity of model reasoning and high deployment costs unfortunately erode fairness and organizational legitimacy. These findings are fully aligned with recent research on the dynamics of human-AI interaction, revealing that the psychological and behavioral impacts of AI integration are far from trivial. For example, Li et al. (2025c) show how employee perceptions of AI-led versus human-led workflows can significantly affect workplace procrastination through mediating factors such as boredom and self-evaluation. Therefore, our review highlights HITL governance as the most common mitigation strategy, yet empirical validation of HITL effectiveness, especially in underrepresented HR functions, is scarce.

The evidence produced by the review also highlights the main answers to the three initial research questions. In particular, RQ1 is answered by bibliometric and functional classification. The retrieved literature shows a clear shift towards generative LLMs, especially in the post-2023 period, while application-oriented papers relying on GenAI principles account for about 52% of the technical/application subset. These deployments focus mainly on resume screening, interview support and adaptive learning, thus leaving performance management, stand-alone HR analytics, compensation and culture largely less developed, signaling both the breadth and the blind spots of real-world or experimental adoption. With respect to RQ2, it is clear that the retrieved literature falls into two dominant build paths: (a) decoder-only models consumed via proprietary APIs and (b) encoder-decoder or encoder-only models that organizations fine-tune in-house, together with recurring engineering patterns, such as prompt design, lightweight fine-tuning and mixed embedding-plus-retrieval pipelines, that steer evaluation choices and performance claims. RQ3 is illuminated by the governance lens layered onto every paper. In particular, most papers suggest a HITL checkpoint or audit step, acknowledging fairness, transparency and bias concerns. However, rigorous tests of HITL effectiveness, data dignity safeguards and liability allocation are still rare, marking these issues as the field’s next empirical frontier. Therefore, the proposed classification not only catalogs the HR tasks LLMs already touch, but also pinpoints the design decisions and oversight mechanisms that must mature for these tools to become both trustworthy and widely diffused.

8.1. Future research

To turn today’s conceptual gaps, open issues and technical limitations into cumulative, decision-ready evidence, we propose a specific road map for future research directions. The strategic road map presented in Fig. 8 can be read as a two-dimensional matrix that locates concrete research actions at the intersection of a time horizon (short-,

Technology trustworthiness pillars

	Data and model robustness	Human-in-the-loop schemes	Governance, ethics & compliance	Autonomy and agentic AI risks
Short-term objectives	Reproducible prompts; multilingual stress tests	Hand-off points mapping in live HR workflows; A/B tests on reviewer's load vs accuracy	Model cards including energy and bias metrics; GDPR-plus audit check-lists; privacy-by-design; prompt-injection tests	Sandbox single-task agents with kill-switches; jailbreak safeguards; action sandboxing
Mid-term objectives	Open, multimodal HR datasets; Domain-specific hallucination metrics; Industry 4.0 interoperability; RAG/blockchain/IoT links	Longitudinal studies with varying HITL depth; Design of dashboards for model uncertainty	Cross-jurisdiction impact assessments; Third-party fairness audits; security audits; poisoning-risk assessment	Verifiable escalation policies in multi-agent stacks; Chain-of-thoughts continuous instrumentation; tool-use monitoring; resume attack isolation
Long-term objectives	Self-healing models with on-line drift detection	Codify "optimal calibration" standards by task criticality; cognitive offloading limits; independent judgment preservation	Harmonization of AI-HRM governance standard (ISO-like); regulated scaling scenario	Agentic HRM transformation; productivity-displacement realignment; human capability protection; misalignment benchmarks

Fig. 8. Advancing the trustworthiness of GenAI applications: A road map for further research in HRM.

medium-, and long-term) and four 'trustworthiness pillars of technology'. In the Data & Model Robustness column the early focus is on reproducible prompt suites and multilingual stress tests that directly plug the data-quality and generalizability holes flagged in today's literature. Mid-term work scales that effort by releasing open, multimodal HR corpora and hallucination-tracking metrics, creating the empirical backbone that current prototypes lack. The long-term ambition is self-healing models with on-line drift detection, closing the performance-and-maintenance gap that now keeps most systems in the lab. In parallel, the Governance, Ethics & Compliance pillar starts with richer model cards (bias, energy, GDPR check-lists) and moves towards cross-jurisdiction impact audits before converging on an ISO-style, AI-HRM governance standard—thereby transforming today's fragmented, reactive compliance practices into a globally harmonized regime. The right half of the matrix tackles autonomy and human oversight. Autonomy & Agentic-AI Risks begins with sandboxed single-task agents and kill-switches, evolves to verifiable escalation policies and continuous chain-of-thought instrumentation for multi-agent stacks, and ultimately widens its lens to macro-labor impacts, misalignment propensity benchmarks, and large-scale HR re-skilling. Finally, the HITL Schemes pillar progresses from live hand-off maps and A/B tests on reviewer load, through longitudinal studies coupled with uncertainty dashboards, and culminates in codified, task-critical "optimal calibration" standards. Together, these four tracks form a synchronized ladder in which each rung remedies a known limitation (e.g., scarce datasets, opaque reasoning, deployment gaps) while advancing the open-issue frontier from isolated fixes towards systemic, field-wide trust mechanisms.

The proposed road map also clarifies the tight coupling between pillars. Short-term reproducibility protocols, for example, are prerequisites for the mid-term fairness audits and escalation logs that sit under Governance and Autonomy. In contrast, the success of long-term self-healing models depends on the calibration norms emerging from the HITL stream. The road map therefore encourages iterative co-evolution. For example, every new dataset feeds alignment dashboards; every governance audit generates labels for misalignment benchmarks; every HITL experiment supplies human-error priors that inform agentic-AI safeguards. By staging these dependencies over time, the road map offers HRM researchers a sequenced research agenda that can be empirically validated at each step. However, remains coherent enough to meet future challenges such as productivity-displacement trade-offs,

cross-border liability, and emergent misalignment in autonomous HR agents.

Apart from serving as a sequenced research agenda, the proposed road map also captures three wider future scenarios of how LLMs may evolve in HRM systems. In the cautious-adoption scenario, organizations mainly use LLMs as decision-support assistants with strict human oversight, whereas research focuses on explainability, auditability and legal compliance. In the regulated-scaling scenario, broader organizational adoption is supported by standardized governance frameworks, cross-jurisdiction audits and interoperable HR analytics infrastructures that enable LLMs to support large-scale recruitment, training and workforce-planning operations. In a more transformative agentic-AI scenario, semi-autonomous and multi-agent HR systems increasingly coordinate recruitment, onboarding, evaluation and employee-support workflows with limited human intervention. This latter route may yield significant productivity gains but also raises concerns about labor displacement, accountability, security, misalignment and the long-term role of human judgment in organizational decision-making. The road map, by framing these alternative trajectories, improves its forecasting logic and suggests that future HRM outcomes will be dependent not only on technical progress, but also on governance choices, institutional adaptation and workforce acceptance.

Having mapped the trustworthiness journey across time horizons and pillars, we can now sharpen the focus to specific research opportunities. We organize them into six interlocking themes, each designed to close a critical evidence gap identified earlier:

- **Robust data foundations and model reliability:** Future research should build open, longitudinal HRM data commons that stitches together multi-year, multi-modal evidence—resumes, vacancy texts, interview videos, internal-mobility records and post-hire outcomes. This shared repository will support a task-grounded "Reasoning Reliability" benchmark of authentic HR scenarios, allowing researchers to probe logical consistency, counterfactual resilience and hallucination rates with domain-specific scripts. The long-term ambition is self-healing, multilingual LLMs that detect distributional drift in real time and patch themselves with minimal human intervention, so reliability scales with deployment scope. Future architectures should also support interoperable Industry 4.0 HR ecosystems, further enabling secure

integration with blockchain infrastructures, IoT-enabled workplace systems, cloud-based HR analytics and retrieval-augmented organizational knowledge bases.

- **Explainability, governance and cross-jurisdiction compliance:** Work is needed to establish global regulatory sand-boxes in which firms pilot HR LLMs under common red-team protocols, audit templates and transparency disclosures. Insights from these trials should flow into richer model cards, GDPR-plus impact audits and procurement check-lists, paving the way for an ISO-style AI-HRM governance standard that harmonizes ethical, legal and energy-use requirements across regions and industries. These governance frameworks should further incorporate privacy-by-design principles, adversarial robustness testing, poisoning-risk assessments and standardized security audits for prompt injection, jailbreak and data-leakage vulnerabilities.
- **HITL optimization and calibration science:** Researchers should Conduct calibrated HITL field experiments that randomize requisitions to AI-only, AI + junior reviewer and AI + senior reviewer pipelines, logging decision times, override rates, fairness metrics and hiring-manager satisfaction for at least six months. Coupled with longitudinal studies of trust and cognitive load, these data will pinpoint the inflection where extra human scrutiny ceases to add value and inform “optimal calibration” guidelines that match hand-off depth to task criticality. Future studies should also examine whether prolonged dependence on AI assistance weakens independent judgment, critical thinking and long-term professional capability development among HR practitioners.
- **Safe autonomy and agentic-AI alignment:** Future research approaches should create a verifiable agent-orchestration sandbox that chains recruitment, learning and compliance agents under formally proven policies, ensuring no high-risk action bypasses human sign-off. Continuous chain-of-thought logging, explicit escalation maps and misalignment-propensity benchmarks (deception, goal-guarding, shutdown resistance) will allow researchers to quantify and mitigate autonomy risks before live deployment. Such infrastructures may further support the transition from narrowly assistive HR systems towards broader agentic-HRM transformation scenarios in which semi-autonomous agents coordinate recruitment, onboarding, compliance and employee-support workflows.
- **Ethical learning and bias mitigation:** Future pipelines should develop debiased imitation-learning pipelines that inject counterfactual data, solicit active-learning corrections and record full lineage metadata, then compare them with vanilla IL across successive retraining cycles. The aim is a reusable “ethical IL” template that any HR domain can adopt, balancing fast domain adaptation with measurable reductions in bias, drift and opacity.
- **Macro-labor impact, sustainability and workforce re-skilling:** Future research should run multi-site “Economic Turing Test” trials that pit autonomous agents against human temps on short-cycle HR tasks across diverse industries, generating head-to-head evidence on quality, compliance and turnaround time. In parallel, instrument pilots, scale-ups and maintenance phases with life-cycle sustainability telemetry—energy use, cloud spend, carbon intensity—for GPT-4 APIs, distilled domain LLMs and hybrid retrieval systems. Combining productivity and carbon data will clarify whether specific AI architectures warrant adoption, taxation or targeted re-skilling programs aimed at smoothing the productivity–displacement trade-off. Such evidence will also help organizations evaluate alternative futures ranging from cautious AI augmentation to regulated large-scale deployment and productivity–displacement realignment across labor markets.

8.2. Limitations

Despite its breadth and transparent protocol, this review has several methodological limitations that warrant caution. The search strategy centered on the main HRM and GenAI terms, so emerging phrases in adjacent fields, such as “algorithmic coaching” in education, may have slipped through, creating selection bias. Because the corpus was mainly based on peer-reviewed work indexed in Scopus, Web of Science, and IEEE Xplore, early evidence found in white papers, pre-prints or vendor reports may be underrepresented. To offset this, however, we have included a significant number of arXiv papers in order to capture all current trends in this cutting-edge domain. In particular, we used them to elaborate on the open issues. Restricting the analysis to English-language publications further narrows the lens and may miss insights from non-Anglophone labor markets with different adoption pathways. Although dual coding and inter-rater checks tempered subjectivity, thematic classifications inevitably reflect the authors’ disciplinary perspectives, and, importantly, no formal risk-of-bias or methodological-quality appraisal of the 141 included studies was performed, leaving the weight of the evidence unranked. Finally, with LLM research accelerating, studies published after the May 2026 cut-off are already reshaping the landscape, underscoring the value of “living” reviews and periodic updates in this fast-moving domain.

9. Conclusions

The evidence in the retrieved literature portrays a field propelled more by excitement than by depth. LLM tools already streamline recruitment, training and first-wave people analytics. However, empirical coverage is narrow, methods uneven and socio-technical risks (like bias, opacity, security and energy cost) slightly covered. By grouping these weaknesses into recurring categories, we show precisely where knowledge is thinnest and why single-shot lab evaluations cannot stand in for longitudinal, context-rich research. The proposed road map for future research directions sheds light on specific next steps: shared HR data commons, calibrated HITL field trials, task-grounded reasoning benchmarks, verifiable agent sand-boxes and aligned audit standards. Coupled with governance check-lists for HR leaders and clearer disclosure requirements for regulators, these measures outline a practical path for turning experimental promise into robust, equitable and strategically meaningful HR practice in the digital era.

CRedit authorship contribution statement

Thomas K. Dasaklis: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Formal analysis, Conceptualization. **Panagiotis G. Giannopoulos:** Writing – original draft, Visualization, Resources, Methodology, Formal analysis, Data curation. **Dimitris Koutras:** Writing – original draft, Resources, Formal analysis, Data curation. **Vangelis Malamas:** Writing – original draft, Resources, Formal analysis, Data curation. **Panos T. Chountalas:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Formal analysis.

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Data availability

Data will be made available on request.

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